

Micromouse

Introducing "Zirconia"

Noriaki

Nakagawa RT Mouse Club

[Micromouse Camp 2024](#)

Presentation materials for June 22, 2024

table of contents

- Self-introduction • Zirconia? • Zirconia concept (as of January 2019) • How to start Zirconia development • Zirconia specs • Zirconia running video 1: Mai's run • Zirconia running video 2: Kohiro's run • Representative Zirconia competition results • Zirconia design information published on Bitbucket • Zirconia sample program coding rules • Zirconia sample program API index.h • Complete set of Zirconia hardware design information • Minimum required program for competitions • Nakagawa's impressions after using Zirconia • Zirconia challenges

self-introduction

- Noriaki

Nakagawa • RT Engineer

- Facebook: <https://www.facebook.com/noriaki.nakagawa.77> •

GitHub: <https://github.com/noriakinakagawa> •

Mouse training blog: [Micromouse training](#)

Company new products,
blog introductions, exhibition
announcements

Power supply
circuit data released

Company-wide homemade mouse training and development reports and
competition participation reports published

Zirconia?

Zirconia, eight-time All Japan MM champion Arumai, will reveal the 2019 hardware design and manufacturing information for the Micro Mouse.

Zirconia is simple hardware, does not require excessive weight reduction or a low center of gravity, and does not require expensive parts. The concept is to create a robot that can be made by anyone who has experience in robotics competitions, such as those who belong to a club or have experience in micromouse robots nearby, by creating a robot using a printed circuit board.

In addition, [Kohiro](#) has written a sample program and device driver, and has also released the above hardware information on Bitbucket under the [Apache License 2.0](#).



Image source: [My Mouse Production Record](#)

Zirconia concept (as of January 2019)

Using the latest technology parts, it is packed with the bare minimum and is simple and easy to use.

1. Latest technology ÿ Updating discontinued parts (gyro: MPU6500 ÿ ICM20468, encoder ÿAS5050ÿMA300) ÿ

Use the DMM 3D printing service (for example, in the case of a wheel with gears, the cutting ratio
Number of parts: 3 ÿ 1, Cost: 1/3)

2. The simplest hardware: motorcycles

The drive structure is simple, and
assembly and disassembly are easy, making maintenance of the
mechanism easy. The electronic circuit is also easy to repair, with a minimum component size of
1608 (the recent trend is 1005), making the MicroMouse easy to use.

Text citation: Hard Manual .pdf

How to start Zirconia development

- Target audience •

Micromouse and other robot competitions, business robots, and those who design and create their own robots.

- Experience •

Have experienced people close to you •

- Assumed starting method •

Obtain design data, make design changes as needed, order parts, and manufacture
do

- Self-manufacturing using mechanical parts and electronic boards with parts mounted

Nakagawa's

example: Complete a commercially available kit (Pi:Co series, HM-Starter
kit, etc.) ÿ Complete a homemade robot race or a homemade classic mouse
ÿ Complete a Zirconia ÿ
Complete a homemade micromouse

Acquire basic robotics skills, acquire
circuit and board design skills, acquire
control-related skills, and
develop a deep world

Zirconia Specs

CPU Clock ///ata	System / micro 32413 40[z] / 1.5[] / 320[k] / 0[k]
battery	CPU: io/ 1 [cell or unit]/ 3.7[]/ 50[mh]/ anggood Motor system: CPU related shared
Drive motor	Motor 2 [pcs] k06-4.5 Motor driver: 8836
Motors other than those for driving	No other No other 2
Sensor	Infrared sensor: 4 [pcs] / 5311+-4206 / Distance sensor (converts reflected light amount) / Gyro sensor: 1 [pc] / 20648 / Digital value output type / Encoder: 2 [pcs] / 300 / Magnetic /
Size: Length x Width x Height [mm] Weight [g]	70x39x13 [mm] 12[g]
driving wheels training wheels	2 wheels/diameter 13mm width 3.5mm Training wheels
Robot structure left and right (2 wheels) degree difference method	
Maximum speed in straight line/acceleration/turning speed	2 [m/s]/ 8 [m/s/s]/ 0.75 [m/s]
Development environment 32ube	:i

Zirconia driving video 1 Maisan driving



Zirconia driving video 2 Kohiro's driving



探索アルゴリズムは未探索優先足立法です。
未探索区間を歩数マップより優先するのでなかなかゴールには辿り着きません。

Zirconia representative tournament results

第36回マイクロマウス東日本地区大会

走行記録

2018.9.1(土)パシフィコ横浜

マイクロマウス競技												
順位	出走	ロボット名	参加者名	グループ名	記録	備考	Point	TR1	TR2	TR3	TR4	TR5
1	26	Fantom2nd		D-The-Star	00:02.138	優勝	10	00:22.557	00:02.676	00:02.537	00:02.411	00:02.138
	27	Zirconia		D-The-Star	00:03.645			00:25.276	00:05.330	00:03.906	00:03.652	00:03.645
2	16	シン・ウムオ		東京理科大学 Mice	00:03.647	第2位	6	R	00:04.512	00:03.869	00:03.647	
3	21	しゅべるま〜じゅにあ		D-The-Star/MiceBusters	00:04.247	第3位	4	00:40.906	00:04.247	00:04.273		
4	24	trane		D-The-Star	00:07.917	特別賞	3	00:42.442	00:08.159	00:07.917	00:07.957	00:08.053
	25	ARROWHEAD		D-The-Star	00:08.844			00:45.278	00:12.718	00:11.680	00:08.844	00:10.878
5	19	neu		Mice Busters	00:10.738	特別賞	2	00:42.168	00:10.738	R	R	R

2020 All Japan Tournament

マイクロマウス競技 エントリー19台

順位	ロボット名/参加者名/所属/動画 URL	BEST	TR1	TR2	TR3	TR4	TR5	備考
1	Fantom3rd(ファントムサード)							
	D-The-Star https://www.youtube.com/watch?v=GgYu0Z9WFrE	00:02.204	00:44.980	00:02.384	00:02.286	00:02.208	00:02.204	特別賞
2	赤い彗星 2.1(アカスイセイ)							
	D-The-Star https://youtu.be/cKaoB16kc0I	00:02.278	00:46.858	00:02.384	00:02.316	00:02.278	00:02.279	特別賞
3	栗もなか(クリモナカ)							
	渋谷教育学園横浜高等学校物理部 https://youtu.be/6-PpZePZTjI	00:02.670	01:00.190	00:02.700	R	00:02.670	R	特別賞
4	x9(エックスナイン)							
	https://youtu.be/mb14m3vj6HM	00:03.342	01:16.751	00:03.342	00:03.730			
5	ロング 19 号機(ロングジョウキョウゴウキ)							
	https://youtu.be/E-OX193TUA	00:03.483	01:03.302	00:03.483				特別賞
6	Thrush(スラッシュ)							
	TRD(トンデモロボット同好会) https://youtu.be/N-0pEOP1QI	00:03.749	00:59.902	00:03.777	00:03.777	R	00:03.749	
7	ZrF4(フッカジルコニウム)							
	D-The-Star https://www.youtube.com/watch?v=6Hd0n26SL0g&feature=youtu.be	00:04.772	01:44.573	00:05.818	00:05.260	00:04.990	00:04.772	

Top ranking:
Complete all runs

Zirconia design information published on Bitbucket

Bitbucket ブルリクエスト リポジトリ プロジェクト

 Zirconia_Sample

[ソース](#)
[コミット](#)
[ブランチ](#)
[ブルリクエスト](#)
[パイプライン](#)
[Deployments](#)
[Jira issues](#)
[セキュリティ](#)
[ダウンロード](#)

kohiro / Untitled project

Zirconia_Sample

master Files Filter files

/

Name	Size	Last commit	Message
hardware		2020-04-13	readmeの追加と細かい修正
software		2023-08-17	最適化対策とリファクタリング
.project	208 B	2022-07-31	フォルダ構成の変更
license.txt	546 B	2022-04-04	ライセンスの追加
readme.md	3.15 KB	2022-07-31	readmeの修正
zirconia.jpg	29.67 KB	2022-07-31	readmeの修正
ソフトマニュアル.pdf	849.04 KB	2020-04-19	マニュアル更新
ハードマニュアル.pdf	3.47 MB	2020-04-13	readmeの追加と細かい修正

URL: https://bitbucket.org/kohiro/zirconia_sample/src/master/

Zirconia Sample Program Coding Rules

"We aim to create a simple and easy-to-use driver that uses the HAL library so that beginners can easily understand and use it." We also put effort into writing sample code.

- Follow naming conventions

For function names, use the convention "filename_verb + ýý", for variable initialization use reset, and for running use run.

- Fit within the screen Write

so that it can be easily scrolled. Make it long enough to be easy to write, but short enough to be reasonably readable.

- File separation is done with header

If you declare index.h, you can use the functions, and if you write instructions and comments, you can see here and use the functions.

- Don't write unnecessary comments

It is best to write code so that it is easy to understand even if you write comments. If you forget to edit a comment, it will cause confusion later.

- Be aware of scope. Make the

code as small as possible. Use global variables that span files and reduce the number of relevant local variables. The number of variables and functions to be managed increases. Try to keep the number of live variables and functions as small as possible, as they are breeding grounds for bugs.

Zirconia sample program API index.h

zirconia_sample / software / Module / Inc / index.h

```

74  /* モータ関数群 (motor.c) */
75  void      Motor_Initialize( void );           // モータ駆動用タイマーの開始
76  void      Motor_StopPWM( void );             // モータを停止
77  void      Motor_SetDuty_Left( int16_t );      // 左モータを指定したDutyで回転させる[0-1000]
78  void      Motor_SetDuty_Right( int16_t );     // 右モータを指定したDutyで回転させる[0-1000]
79
80  /* エンコーダ関数群 (encoder.c) */
81  void      Encoder_Initialize( void );         // エンコーダ用タイマーの開始
82  void      Encoder_ResetCount_Left( void );    // 左エンコーダのカウントを初期値にする
83  void      Encoder_ResetCount_Right( void );   // 右エンコーダのカウントを初期値にする
84  float     Encoder_GetAngle_Left( void );      // 左タイヤの角度を取得する[rad]
85  float     Encoder_GetAngle_Right( void );     // 右タイヤの角度を取得する[rad]
86
87  /* 慣性センサ関数群 (imu.c) */
88  uint8_t   IMU_CheckWHOAMI( void );           // 慣性センサの動作確認回数 (0x0が返ってくれば正)
89  void      IMU_Initialize( void );            // 慣性センサの初期設定
90  void      IMU_ResetReference( void );        // 慣性センサのリファレンスを補正する
91  float     IMU_GetAccel_X( void );            // X軸加速度計の加速度を取得する[m/s^2]
92  float     IMU_GetGyro_Z( void );            // Z軸ジャイロの角速度を取得する[rad/s]
93
94  /* 赤外センサ関数群 (ir_sensor.c) */
95  void      Sensor_Initialize( void );         // AD変換の初期設定
96  void      Sensor_StartADC( void );          // AD変換を開始する
97  void      Sensor_StopADC( void );           // AD変換を停止する
98  uint16_t  Sensor_GetBatteryValue( void );    // 電源電圧のAD値を取得する
99  int16_t   Sensor_GetValue( uint8_t );       // 赤外センサのLEDオンオフ差分値を取得する
100
101  void      Sensor_DebugPrintf( void );
102
103  /* バッテリー関数群 (battery.c) */
104  float     Battery_GetVoltage( void );        // バッテリーの電圧を取得する[V]
105  void      Battery_LimiterVoltage( void );    // バッテリーの電圧が3.2V以下になると起動しないように制限する
106
107  /* 割り込み関数群 (interrupt.c) */
108  void      Interrupt_Initialize( void );      // メインの割り込み処理の初期設定
109  void      Interrupt_Main( void );           // メインの割り込み処理を書く
110  uint16_t  Interrupt_GetDuty( void );        // 割り込み処理内の計算割合を取得する
111  uint16_t  Interrupt_GetDuty_Max( void );     // 割り込み処理内の最大計算割合を取得する
112  float     Interrupt_GetBootTime( void );     // マイコンが起動してから経過した時間を取得する[s]
113
114  /* フラッシュ関数群 (flash.c) */
115  uint8_t   Flash_Clear( void );              // Flashのデータを消去
116  uint8_t*  Flash_Load( void );              // Flashのデータを全てに読み出し先頭アドレスを取得する
117  uint8_t   Flash_Save( void );              // Flashにデータを書込む
118
119  /* モジュールテスト関数群 (module_test.c) */
120  void      module_test( void );              // 全モジュールの動作確認用テスト関数
121
122  #endif /* INC_INDEX_H */

```

URL: https://bitbucket.org/kohiro/zirconia_sample/src/master/software/Module/Inc/index.h

Complete set of Zirconia hardware design information

zirconia_sample / hardware

Name	Size	Last commit
↑ ..		
gerber		2020-04-13
Zirconia-all-pos.csv	4.37 KB	2020-04-13
Zirconia-all-pos.xlsx	12.7 KB	2020-04-13
Zirconia_BOM_PCBGOGO.xlsx	3.32 MB	2020-04-13
Zirconia_BOM_fusionPCB.xlsx	12.9 KB	2020-04-13
Zirconia_CircuitData_KiCAD.zip	816.83 KB	2020-04-13
Zirconia_circuit.pdf	101.16 KB	2020-04-13
gerber.zip	61.04 KB	2020-04-13
マウス部品リスト.xlsx	17.15 KB	2020-04-13

Gerber data PCB

mounting parts list.csv

PCB mounting parts

list.xlsx PCBgogo mounting request

data FusionPCB mounting request

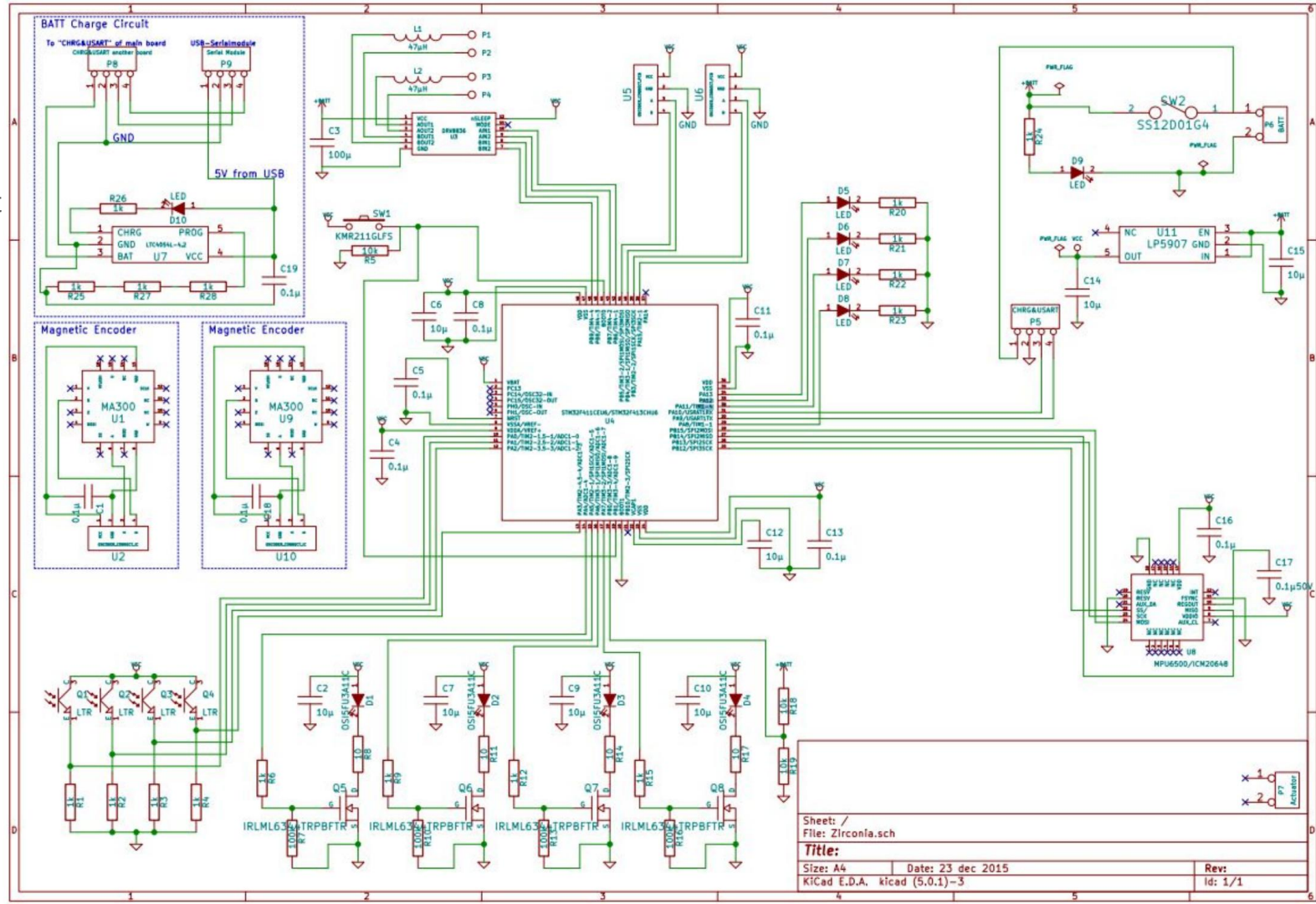
data KiCad

data

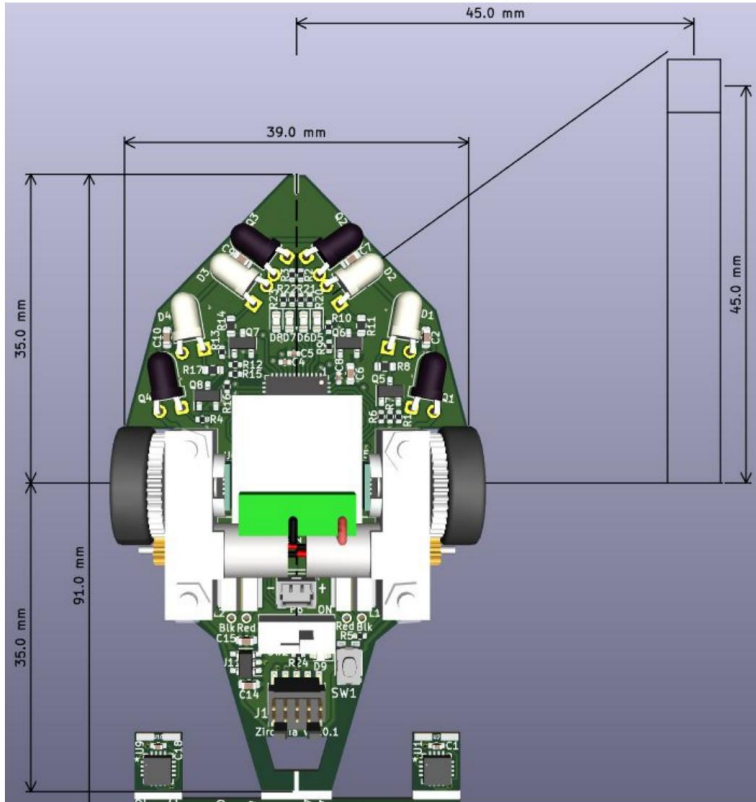
Circuit diagram Ordering

Gerber data zip Parts list

Zirconia(v.1.0) circuit diagram



Sensor placement explanation



"When using a mouse with a sensor that can see half a section ahead, five turns (large diameter 90 degrees, 180 degrees, 45 degrees, 135 degrees, and diagonal 90 degrees) can be combined to make diagonal turns, and wall-break correction is possible just before the turn. There are no exceptions to this rule, even in maze patterns.

Therefore, it is not possible to make a turn just before the wall. The mechanism to prevent this is effective.

Text quoted: [Robot Craft Laboratory Wall Cut Correction F Note](#)

[Does the Laughing Gate also have wall cuts?](#)

Minimum program required for the tournament

1. Wall presence/

absence determination 2.

Mode switching

execution 3. Log saving 4. Degree/

angle control: a. Wheel degree estimation, body

degree/angle estimation b. Feedback control (PID, etc.)

5. Moving forward one section: Position control (trajectory generation,

acceleration setting) 6. Pivot turn: Angle control (trajectory generation, angle

acceleration setting) 7. Wall

distance attitude control 8. Maze algorithm

implementation: Left method 9. Maze

information utilization: Foot method 10. Shortest running

Nakagawa's impressions after using Zirconia

- The reference machine hardware is complete, so try out various controls.

Easy to use maze program debug tool

- The aircraft has a good balance of center of gravity for running, and it is stable and controllable up to the limit (non-suction), so it can exceed the limit (adding several Gs) when transplanting your own aircraft.
- Use this machine to make your own machine (suction machine) as soon as possible.

Zirconia Challenge

- Parts are out of stock and out of stock on the market
- Slow writing speed
- Display LED is hard to see
- Communication connector is easily damaged when plugged in and unplugged
- Push button usage limit
- LED malfunction for measuring distance to left wall
- Magnetic encoder linearity is poor

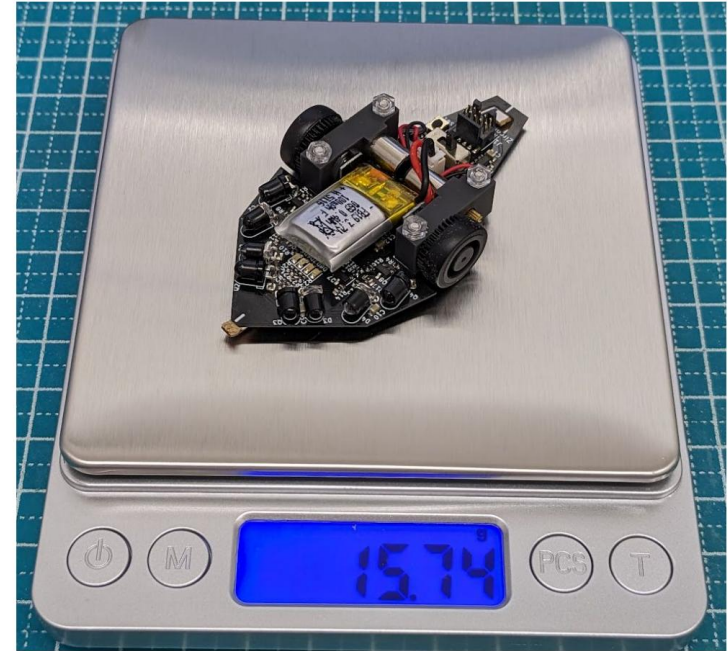
Introducing Zirconia v.2.0

table of contents

- Changes in Zirconia v.2.0 • Zirconia v.2.0 specifications • Zirconia v.2.0 features • Zirconia v.2.0 design information published on Bitbucket • Bill of materials 1: Mechanical and DIP parts • Bill of materials 2: SMD parts • Zirconia v.2.0 circuit diagram A • Zirconia v.2.0 circuit diagram B • 3D STEP data (including mechanical) in the KiCad folder is also included • JLCPCB surface mounting requests are assumed from the start • JLCPCB parts and mounting costs (when 5 boards are mounted) • PCBgogo board mounting • Compatible with Zirconia v.1.0 chassis • Looking for Zirconia children

Changes in Zirconia v.2.0

- Zirconia v.1.0 problem solved
 - Change discontinued products to make them more readily available
 - Avoid ports that are high during boot mode
 - Display LED position
 - Brightness of green LED for display (resistor)
 - Organize board wiring
- (refactoring) • Reduce mounting and processing costs
- All mounted components on the surface (IMU components)
- Change charging IC to a low-cost product
- Change power connector to PicoBlade
- Add ST-LINK
 - Increased writing efficiency
 - Reduced push button usage frequency
 - Enables debugging function
- Communication connector insertion/removal
- Increased battery charging efficiency
- Constant current 50mA ÷ 100mA
- Improved measurement quality with magnetic encoders
 - Magnet with increased magnetic flux density
 - Motor mount magnetic encoder magnet positioning jig function added
 - High resolution sensor IC
 - 3D printed gear to machined gear (return)
 - Placement due to board wiring
 - Power ON/OFF using 3V regulator EN pin
 - Motor capacitor 100uF ÷ 20uF



Board weight increased by 0.98g Battery

weight increased by approximately 1g

Undercarriage weight increased by approximately 2g

Zirconia v.2.0 specs

CPU Clock ///ata	System / micro 32411 100[z] / 512 [k] / 128 [k] / 0[k]	Down
battery	CPU: io/ 1 [cell or unit]/ 3.7[]/ 100[mh]/ Unknown Motor system: CPU related shared	UP
Drive motor	Motor 2 [pcs] -0612 Motor driver: 8836	
Motors other than those for driving	There is no other No other 2	
Sensor	Infrared sensor: 4 [pcs] / 5311+-4206 / Distance sensor (converts reflected light amount) / Gyro sensor: 1 [pc] / -42688- / Digital value output type / Encoder: 2 [pcs] / 730 / Magnetic /	UP
Size: Length x Width x Height [mm] Weight [g]	70x39x13 [mm] 15.7[g]	Down
driving wheels	2 wheels/diameter 13mm width 3.5mm	
training wheels	Training wheels	
Robot structure left and right (2 wheels) degree difference method		
Maximum straight line speed/acceleration/turning speed [m/s]/ [m/s/s]/ [m/s]		
Development environment 32ube	:i	

Zirconia v.2.0 Features

- Magnetic encoder with high resolution and improved linearity eliminates the need for a moving average filter (multiple

minutes)

- Fast parameter adjustment with writing • Increased battery capacity for longer

adjustment time and running time • Easy adjustment of center distance by adjusting cutting gear meshing

(no need for gear break-in)

(required)

Zirconia v.2.0 design information published on Bitbucket

workspace / nnakagawa

Zirconia_Sample

Fork and branch

Here's where you'll find this repository's source files. To give your users an idea of what they'll find here, [add a description to your repository](#).



v.2.0 ▾

Files ▾

Filter files



/

ÿ

Name

Size

Last commit

Message



hardware

3 日前

update



software

3 日前

add AE-FT234X



.project

208 B

2022-07-31

フォルダ構成の変更



license.txt

549 B

2024-06-02

first commit



readme.md

2.86 KB

2024-06-02

first commit



zirconia_v.2.0.jpg

55.93 KB

2024-06-02

first commit



ソフトウェアマニュアル.pdf

849.04 KB

2020-04-19

マニュアル更新



ハードウェアマニュアル.pdf

3.47 MB

2020-04-13

readmeの追加と細かい修正

URL: https://bitbucket.org/nakagawanoriaki/zirconia_sample/src/v.2.0/

How to download Zirconia v.2.0 branch

▼ ブルリクエスト リポジトリ プロジェクト

Q Search

C Zirconia_Sample

- ソース
- コミット
- ブランチ
- ブルリクエスト
- パイプライン
- Deployments
- Jira issues
- セキュリティ
- ダウンロード**

workspace / nnakagawa / Zirconia_Sample

ダウンロード

i For large uploads, we recommend using the API. [Get instructions](#)

ダウンロード タグ **ブランチ**

ブランチ	コミット	日付	ダウンロード
v.2.0	e968368	2024-06-15	zip · gz · bz2
master	c92f9ac	2023-08-17	zip · gz · bz2

Parts List 1 Mechanical and DIP parts

大分類	品目	品名/型番	入手先	個数	参考URL	参考価格(税込)
機構部品	ビニオンギア	精密切削ビニオンギヤ(m0.3 T9 2mm) 2個入り	RT	2	https://www.rt-shop.jp/index.php?main_page	880
	モーター	コアレスモータ NFP-D0612 (2個入)	RT	2	https://www.rt-shop.jp/index.php?main_page	1100
	タイヤ	マイクロマウス用タイヤ10度 10個入り	RT	2	https://www.rt-shop.jp/index.php?main_page	2200
	カグスパール	ニチアス カグスパール フリーサイズ 1シート 茶 321230	amazon	1	https://amzn.asia/d/05YCirS	660
	磁石	ネオジム磁石(N40)、円柱型、Φ3x2(mm)、径方向	ネオマグ	2	https://www.neomag.jp/shop/shoppingcart/it	98
	3Dプリンタ部品	DMMの3Dプリントサービス 材質は高精度プラスチック	DMM make	1	https://make.dmm.com/mypage/upload/	3100
	平ギヤ	Gearwheel Pinion M0.3 / 38teeth OD 12mm drill3mm、width1.5、Polyacetal POM black	kkpmo	2	https://shop.kkpmo.com/product_info.php?i	2572
	ベアリング	ベアリング外径6内径3厚2.5 MR63ZZ	モノタロウ	4	https://www.monotaro.com/p/8892/7027/	876
	ネジ	(+)スリムヘッド小ねじ (ステンレス)(/パック品)	モノタロウ	4	https://www.monotaro.com/p/4174/6731/74	450
	ナット	M2 六角ナット(ポリカーボネート/10個)	モノタロウ	4	https://www.monotaro.com/g/02444870/74	495
	平行ピン	3x8 平行ピン 硬質(ステンレス)	モノタロウ	2	https://www.monotaro.com/p/4220/5283/74	439
	シムリング	3x4mm セットアップ・ステンレスシム 0.1/0.2/0.3mm厚 各10枚入 [EG-SH3]	スーパーラジコン	1	https://super-rc.co.jp/rc/product/view?id=30	347
未実装電子部品	バッテリー(4つのうちどれか1つでOK)	100mAhリチウム電池モデルバッテリー-SYMA X12S用	amazon		https://amzn.asia/d/7ZSeMQ6	277
	バッテリー(4つのうちどれか1つでOK)	Tygsz M1 KFPLAN KF606蒸気RC飛行機のための1S 3.7V 50mAhミニリポ電池モレックス 2P 1.25mmコネクタ1cm 4g	Banggood	1	https://jp.banggood.com/1S-3_7V-50mAh-Li	
	バッテリー(4つのうちどれか1つでOK)	100mAh ミニRCクワッドコプター用	Aliexpress	1	https://ja.aliexpress.com/item/10050025965	
	バッテリー(4つのうちどれか1つでOK)	3.7V 100mAh リポバッテリー	amazon	1	https://amzn.asia/d/j8jUXm	
	バッテリーコネクタ(P6)	PicoBlade 530470210	Digikey	1	https://www.digikey.jp/short/10232fr3	41.8
	磁気式エンコーダ(U1, U9)	MA730GQ-Z or MA302GQ-Z	Digikey	2	https://www.digikey.jp/short/b3m30fzz	2076.8
	ST-LINK(U12)	STLINK-V3MODS	Digikey	1	https://www.digikey.jp/short/mr74wj20	1444
	壁センサ受光素子(Q1~4)	LTR-4206E	Digikey	4	https://www.digikey.jp/product-detail/ja/lite-on-in	281.6
	ピンヘッダ2x5(J1, J4)	ピンヘッダ 2x5(10P) 表面実装用 1.27mmピッチ	akiduki	2	https://akizukidenshi.com/catalog/g/g11090	60
	壁センサLED(D1~4)	OSI5FU3A11C	akiduki	4	http://akizukidenshi.com/catalog/g/g1-04313/	200
	電源スイッチ(SW2)	SS12D01G4	akiduki	1	http://akizukidenshi.com/catalog/g/gP-1272/	25
	充電、通信ハーネス(P8)	2x5(10P)両端コネクタ付IDCリボンケーブル(フラットケーブル ハーフピッチ)	akiduki	1	https://akizukidenshi.com/catalog/g/g10677/	240
ハーネス	USBケーブル microB	通信ができるタイプなら何でもOK	akiduki	1	https://akizukidenshi.com/catalog/g/g11701/	90
その他準備品	ドライバ0番					
	はんだごて(20~30W)、先が細いもの					
	ゼロハンテープ、できればカプトンテープ					
	エポキシ系接着剤	例えば、コニシのボンドクイック5				
	アロンアルファ					

Parts list 2 SMD parts

品名 (Name)	部品番号 (Designator)	個数 (Quantity)	値 (Value)	パッケージ (Package)	説明 (Description)	部品メーカー (Manufacturer)	部品メーカーの型番 (Manufacturer Part Number)
セラミックコンデンサ	C1, C4, C5, C8, C11, C13, C16, C18, C19	9	0.1uF 16V	表面実装1005mm		Samsung Electro-Mechanics	CL05B104KP5NNNC
セラミックコンデンサ	C2, C6, C7, C9, C10, C12, C14, C15, C20, C21	10	10uF 25V	表面実装1608mm		Samsung Electro-Mechanics	CL10A106MA8NRNC
セラミックコンデンサ	C17	1	10nF	表面実装1005mm		Murata Electronics	GCM155R71H103KA55D
LED	D1, D3	2	OSI5FU3A11C		向きがあるので、実装注意 (本ファイル別タブ「部品実装向き」参照)	OptoSupply	OSI5FU3A11C
LED	D2, D4	2	OSI5FU3A11C	D1,D3と曲げ向き逆	向きがあるので、実装注意 (本ファイル別タブ「部品実装向き」参照)	OptoSupply	OSI5FU3A11C
LED	D5	1	Yellow	表面実装1608mm	向きがあるので、実装注意 (本ファイル別タブ「部品実装向き」参照)	Broadcom Limited	XL-1608UYC-06
LED	D6, D10	2	Red	表面実装1608mm	向きがあるので、実装注意 (本ファイル別タブ「部品実装向き」参照)	Broadcom Limited	XL-1608SURC-06
LED	D7	1	Green	表面実装1608mm	向きがあるので、実装注意 (本ファイル別タブ「部品実装向き」参照)	Broadcom Limited	XL-1608SYGC-06
LED	D8, D9	2	Blue	表面実装1608mm	向きがあるので、実装注意 (本ファイル別タブ「部品実装向き」参照)	Broadcom Limited	XL-1608UBC-04
コネクタ	J1, J4	2	PH127-2x5MG-1.5		向きがあるので、実装注意 (本ファイル別タブ「部品実装向き」参照)	Useconn Electronics Ltd.	PH127-2x5MG-1.5
インダクタ	L1, L2	2	47uH	表面実装4.00mm x 4.00mm		cjiang	AMWPH4018S470MT
コネクタ	P6	1	PicoBlade 530470210		向きがあるので、実装注意 (本ファイル別タブ「部品実装向き」参照)	Molex	530470210
フォトトランジスタ	Q1, Q2	2	LTR		向きがあるので、実装注意 (本ファイル別タブ「部品実装向き」参照)	Lite-On Inc.	LTR-4206E
フォトトランジスタ	Q3, Q4	2	LTR	Q1,Q2と曲げ向き逆	向きがあるので、実装注意 (本ファイル別タブ「部品実装向き」参照)	Lite-On Inc.	LTR-4206E
MOSFET	Q5, Q6, Q7, Q8	4	IRLML6344	SOT-23-3		Infineon Technologies	IRLML6344TRPBF
MOSFET	Q9	1	DMC2400UV-7	SOT-563	向きがあるので、実装注意 (本ファイル別タブ「部品実装向き」参照)	Diodes Incorporated	DMC2400UV-7
抵抗	R1, R2, R3, R4, R6, R9, R12, R15, R20, R21, R23, R24, R26, R27, R28	16	1k Ohms	表面実装1005mm		YAGEO	RC0402FR-071KL
抵抗	R5, R18, R19, R25	3	10k Ohms	表面実装1005mm		Samsung Electro-Mechanics	RC1005F103CS
抵抗	R7, R10, R13, R16	4	100k Ohms	表面実装1005mm		YAGEO	RC0402FR-07100KL
抵抗	R8, R11, R14, R17	5	10 Ohms 0.25W	表面実装1608mm		Rohm Semiconductor	ESR03EZPF10R0
抵抗	R22	1	510 Ohms				
スイッチ	SW1	1	KMR211GLFS			C&K	KMR211GLFS
スイッチ	SW2	1	SS12D01G4			Switronic Industrial Corp.	SS12D01G4
IC	U1, U9	2	MA730	QFN-16 (3mmx3mm)	向きがあるので、実装注意 (本ファイル別タブ「部品実装向き」参照)	Monolithic Power Systems Inc.	MA730GQ-Z
IC	U3	1	DRV8836DSSR	12-WDFDN	向きがあるので、実装注意 (本ファイル別タブ「部品実装向き」参照)	Texas Instruments	DRV8836DSSR
IC	U4	1	STM32F411CEU6/STM32F413CHU6	48-UFQFN	向きがあるので、実装注意 (本ファイル別タブ「部品実装向き」参照)	STMicroelectronics	STM32F411CEU6
IC	U7	1	MCP73831	SOT-23-5		Microchip Technology	MCP73831T-2ACI/OT
IC	U8	1	ICM-42688-P			TDK InvenSense	ICM-42688-P
IC	U11	1	LP5907MPX-3.0	SOT-23-5		Texas Instruments	LP5907MPX-3.0/NOPB
IC	U12	1	STLINK-V3MOS			STMicroelectronics	STLINK-V3MODS

PCB Mounting Components Digi-key My Lists

[製品](#)
[メーカー](#)
[リソース](#)
[見積もり作成](#)

6,000円以上のご注文の送料は無料!

[ダッシュボード](#)
[部品リスト](#)
[Zirconia v2.0](#)

価格カリキュレーターによってコストを節約できる部品が9件検出されました。 [オプションを確認](#)

Zirconia v2.0

リスト設定

仕入率を定める

① ¥9,679

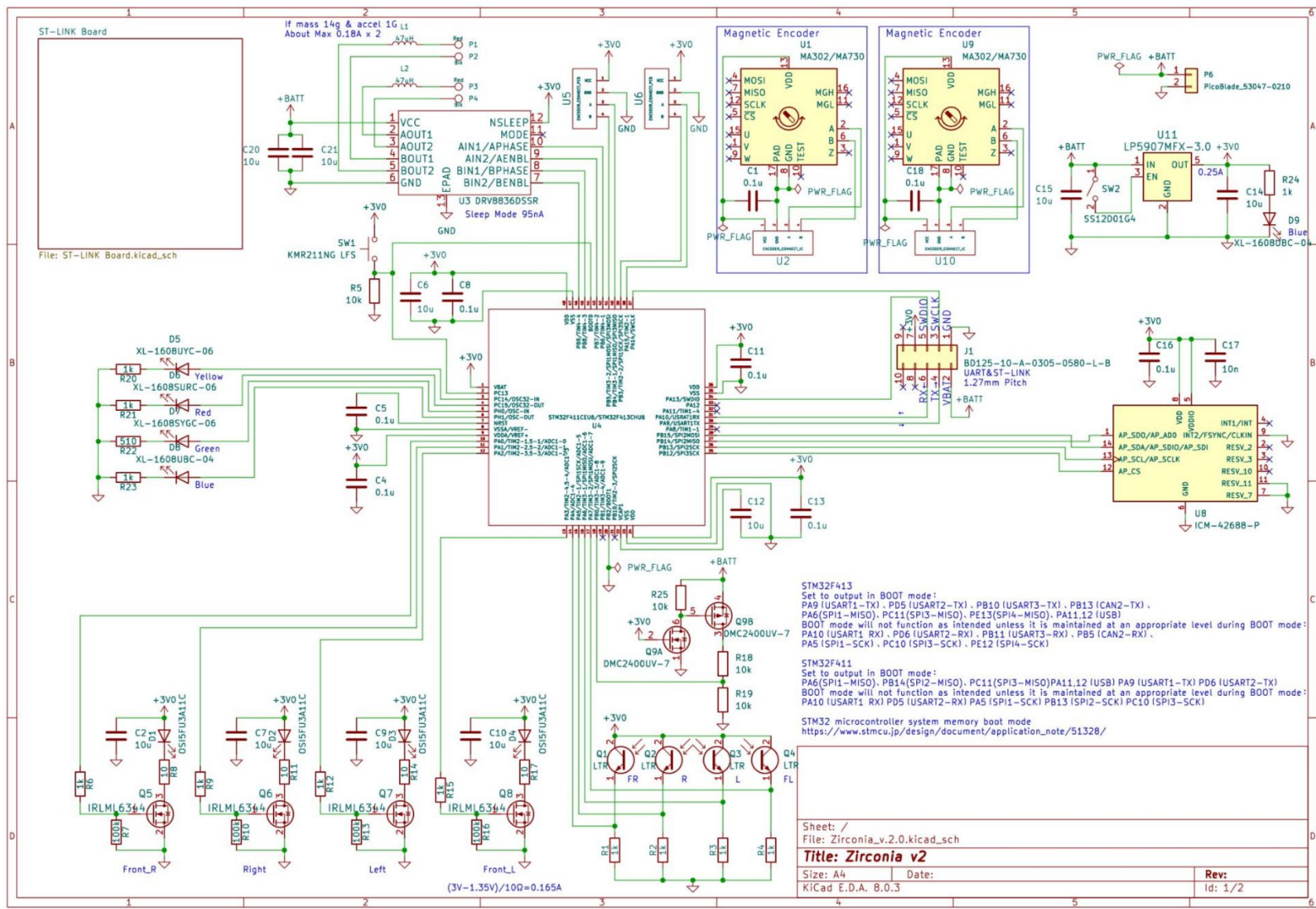
アセンブリ

+	#	部品	入手性	数量	梱包形態	単価	合計金額	最先手部品番号	リクエスト品番	仕税率	最小発注数量
+	1	CL05B104KPSNNVC CAP CER 0.1uF 10V X7R 0402 Samsung Electro-Mechanics	32,262,555 在庫品	9 数量を追加 数量を増やすと、合計金額が増えることができます。 オプションを表示	9 Cut Tape (CT) (1276-1002-1-ND) ~	17.00000	¥153	C1, C4, C5, C8, C11, C13, C1	CL05B104KPSNNVC	%	Cut Tape (CT) : 1
+	2	CL10A106MABNRC CAP CER 10uF 25V X5R 0603 Samsung Electro-Mechanics	1,156,480 在庫品	10 数量を追加 数量を増やすと、合計金額が増えることができます。 オプションを表示	10 Cut Tape (CT) (1276-1889-1-ND) ~	28.40000	¥284	C2, C6, C7, C9, C10, C12, C1	CL10A106MABNRC	%	Cut Tape (CT) : 1
+	3	GCM155R71H100KA55D CAP CER 10000PF 50V X7R 0402 Murata Electronics	3,725,960 在庫品	1 数量を追加 数量を増やすと、合計金額が増えることができます。 オプションを表示	1 Cut Tape (CT) (490-4762-1-ND) ~	17.00000	¥17	C17	GCM155R71H100KA55D	%	Cut Tape (CT) : 1
+	4	HSMY-C190 LED YELLOW DIFFUSED CHIP SMD Broadcom Limited	227,393 在庫品	1 数量を追加 数量を増やすと、合計金額が増えることができます。 オプションを表示	1 Cut Tape (CT) (516-1424-1-ND) ~	67.00000	¥67	D5	HSMY-C190	%	Cut Tape (CT) : 1
+	5	HSMG-C190 LED GREEN DIFFUSED CHIP SMD Broadcom Limited	64,567 在庫品	2 数量を追加 数量を増やすと、合計金額が増えることができます。 オプションを表示	2 Cut Tape (CT) (516-1422-1-ND) ~	67.00000	¥134	D6, D10	HSMG-C190	%	Cut Tape (CT) : 1
+	6	HSMG-C190 LED GREEN DIFFUSED CHIP SMD Broadcom Limited	225,285 在庫品	1 数量を追加 数量を増やすと、合計金額が増えることができます。 オプションを表示	1 Cut Tape (CT) (516-1425-1-ND) ~	62.00000	¥62	D7	HSMG-C190	%	Cut Tape (CT) : 1

[フィードバック](#)
[サポートが必要ですか?](#)

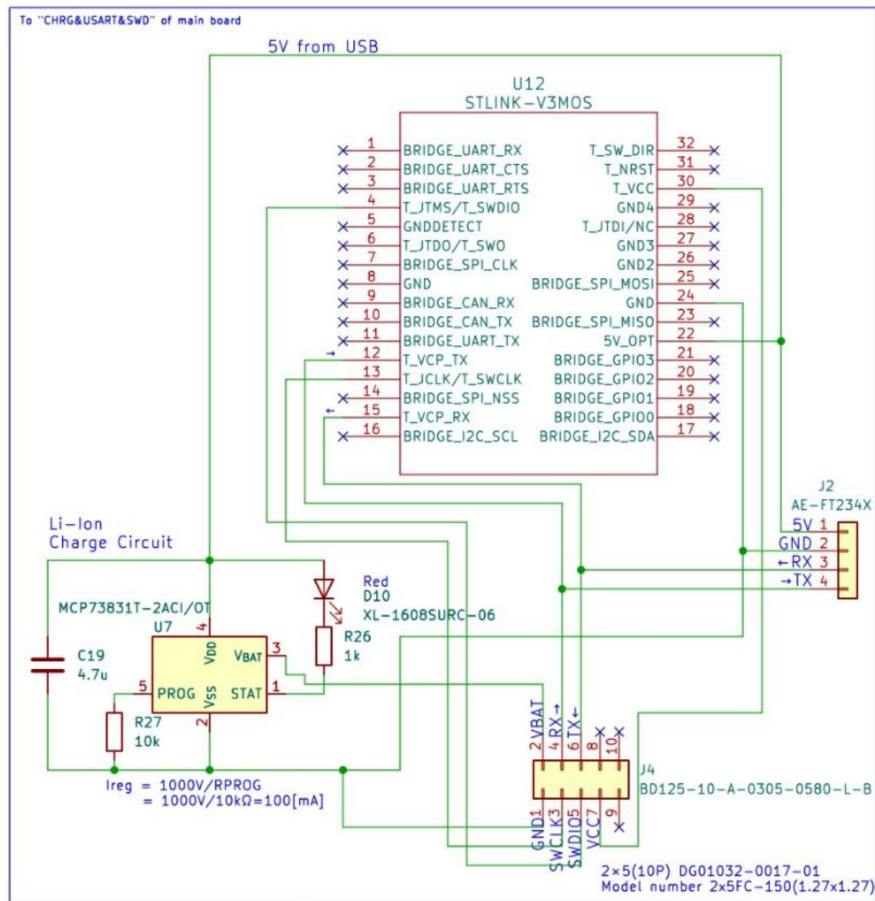
Zirconia v.2.0

Schematic A

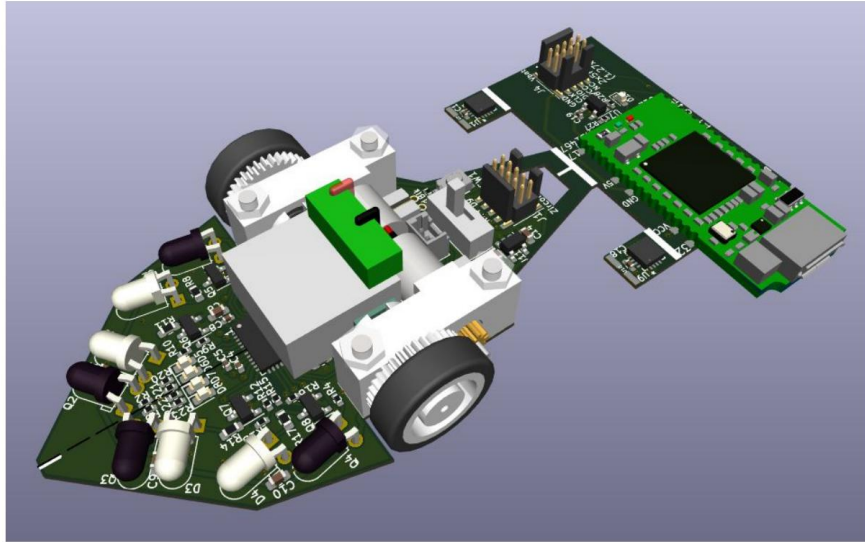


Zirconia v.2.0

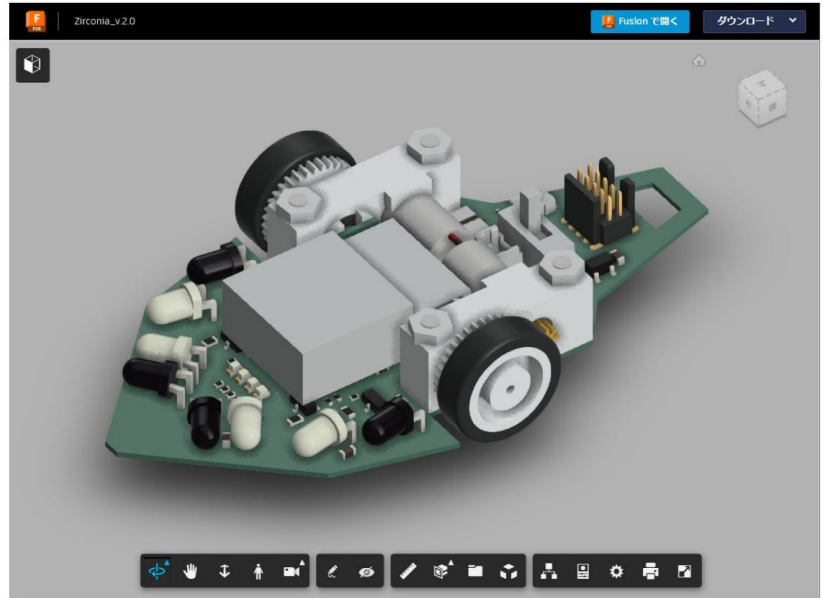
Schematic B



3D STEP data (including mechanical parts) is also included in the KiCad folder.



When making your own, you can reuse each part.



3D data also available online.

[*KiCad Libraries License CC-BY-SA 4.0](https://a360.co/4cjsH3n)

<https://a360.co/4cjsH3n>

Assume JLC PCB surface mounting request from the beginning

Zirconia_v2.0

PCB 部品表 コンポーネントの配置 見積り & 発注

自動的に保存され、最終更新日は 26 May, 17:25 です

2D 3D Simplify Model Top Bottom

☐ Select all top parts

* The preview for reference only. Check final part placement at DFN. Try in 4-6 hours after placing the order.

X: 126.13mm
Y: 24.58mm

2x5FC-150 (1.27x1.27)

トップ・デジグネーター	JLCPCB 部品番号	コメント	フットプリント
C1,C11,C13,C16...	C1525	0.1u	0402
C10,C12,C14,C1...	C19702	10u	0603
C17	C15195	10n	0402
C19	C23733	4.7u	0402
D5	C965802	Yellow	0603
D7	C965805	Green	0603
D8,D9	C965807	Blue	0603
D10,D6	C965799	Red	0603
L1,L2	C2846193	47uH	Inductor
Q5,Q6,Q7,Q8	C53550	IRLML63...	SOT-23
Q9	C177025	DMC240...	SOT-563
R1,R12,R15,R2,R...	C11702	1k	0402
R10,R13,R16,R7	C25741	100k	0402
R11,R14,R17,R8	C22859	10	0603
R18,R19,R25,R27...	C25744	10k	0402
R22	C25123	510	0402
SW1	C221675	KMR211...	KMR211G

次

JLCPCB standard stock

The following SMD components
are manually
mounted: MA730, 2x5
pin header, STLINK-V3MOS

Other DIP products require separate mounting

JLCPCB parts and mounting costs (5 pieces mounted)

Zirconia_v.2.0

PCB

部品表

コンポーネントの配置

見積り & 発注

自動的に保存され、最終更新日は 2 June, 15:44 です

料金詳細

PCB価格	\$15.41
技術料:	\$4.00
ビア処理:	\$0.00
パネル:	\$7.71
表面仕上げ:	\$1.20
ボード:	\$2.50

標準 PCBA の価格

設定料金:	\$25.00
ステンシル:	\$7.86
パネル:	\$42.86
Lサイズ:	\$0.00
構成部品 (22 アイテム):	\$42.10
フィーダー積み込み料:	\$27.66
SMTアセンブリ:	\$2.20
梱包料:	\$0.47

製造時間:

PCB: 24 hours	\$0.00
組み立て: ☒ 2-3 days	\$0.00
☐ 1-2 days	\$47.14

合計価格:

\$163.56

重量:

410.75g

商品説明:

選択してください

カートに保存

\$163.56

25,721.85 yen/5

sheets + Customs fee

invoice from

OCS

2,600 yen ÷ 5,196 yen/sheet

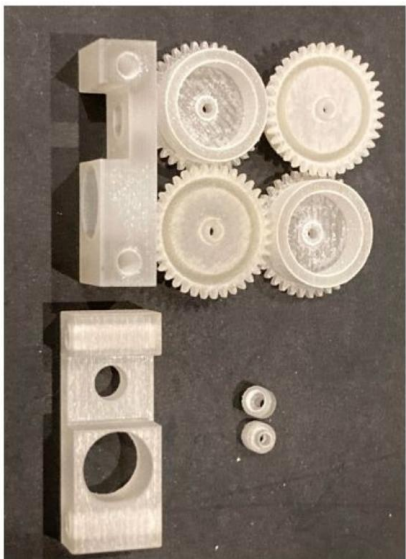
PCBgogo PCB Mounting

部品実装サービス					
注文番号:	S-B83311ASX13	リードタイム:	合計:	\$317.00	49558円
関連の基板注文番号:	B83311ASX12	登録時間:	2024/2/4 19:16:08	製造完了予定日:	2024-03-20 中国時間(GMT + 8)
BOM:	Zirconia_BOM_PCBGOGO.xlsx				
ガーバーデータ:	B83311ASX12_Zirconia.kicad_pcb.zip				
座標データ:	Zirconia-top-pos.csv				
部品調達方法	PCBGOGOより部品を代理購入する	枚数	5	部品点数 (一枚分)	80
SMT点数 (一枚分)	276	DIP点数 (一枚分)	21	面付け方法	面付けなし
実装面	表面	X線検査	なし	購買発注書の注文番号	
ご要望、連絡事項等					
リピート注文履歴					

価格	送料	割引	会員割引	クレジットカード手数料	総額
\$386.00 60345円	\$27.00 4221円	-\$0.00 -0円	-\$0.00 -0円	\$22.00 3439円	\$435.00 68006円

+ Import duty from OCS: 4,800 yen Total: 72,912 yen/5 pieces
y
14,582 yen per ticket

Zirconia v.1.0 suspension compatible



オープンソースのハーフサイズマイクロマウス「Zirconia」の機構部品1台分です。
ホイールは4つありますが、4輪ではありません。
軸ブレが少ない、噛み合わせの良いもの2つを選別して、お使いください。

DMMpartsZirconia.stl

by Mouser_mat

[サイズ] X : 56.687mm × Y : 8.162mm × Z : 36.163mm

造形販売 4,831円～

種類 :



高精細アクリル | MJT | 3500HDMax

発送目安: 3日～10日

▶ 返べる素材・種類について

即納サービスに関する注意事項

- 注文時は必ず即納サービス対象の素材のみで注文をお願いします。
- 通常納期素材と即納サービス素材を混同しての注文はキャンセルとさせていただきます。
- 受付対象曜日及び時刻: 毎週月・火・水・木曜日の15時までの注文については、当日を注文日とします。
- 3Dデータが造形可能なデータであることが受付条件となります。
- 月～水曜日15時以降の注文については、翌日を注文日とします。
- 木曜日15時以降の注文は、翌週月曜日を注文日とします。
- 造形不良が発生した場合、即納できない可能性があります。その際は、当社よりご連絡させていただきます。
- なにかご不明な点や質問がございましたら、お気軽にお問い合わせください。

数量 :

1

価格 :

4,831円



バスケットに入れる

♡ お気に入りに追加

0

✕ ポスト

👍 いいね!

シェアする

Lightweight
people, please

Approximately 2g lighter

URL: <https://make.dmm.com/item/918432/>

We are looking for Zirconia children

Want to try developing with Zirconia?

Members who wish to exchange information and seek advice

We are looking for people to join us.

Development materials reference

Pin talk in boot mode

STM32F413

Boot mode output settings:

PA9(USART1-TX), PD5(USART2-TX), PB10(USART3-TX), PB13(CAN2-TX),
PA6(SPI1-MISO), PC11(SPI3-MISO), PE13(SPI4-MISO), PA11&PA12(USB)

Maintain proper levels in BOOT mode so BOOT mode functions as intended:

PA10 (USART1 RX), PD6 (USART2-RX), PB11 (USART3-RX), PB5 (CAN2-RX),
PA5 (SPI1-SCK), PC10 (SPI3-SCK), PE12 (SPI4-SCK)

STM32F411

Boot mode output settings:

PA9(USART1-TX), PD6(USART2-TX), PA6(SPI1-MISO), PB14(SPI2-MISO), PC11(SPI3-MISO), PA11&PA12(USB)

Maintain proper levels in BOOT mode so BOOT mode functions as intended:

PA10 (USART1 RX), PD5 (USART2-RX), PA5 (SPI1-SCK), PB13 (SPI2-SCK), PC10 (SPI3-SCK)

Reference material: STM32 microcontroller system memory boot mode https://www.stmcu.jp/design/document/application_note/51328/

Left wall distance measurement LED (PA6) output setting during boot mode

AN2606

STM32F413xx/423xx デバイスのブートローダ

表 71. システムメモリブートモードでの STM32F413xx/423xx の設定 (続き)

ブートローダ	機能/ペリフェラル	状態	コメント
SPI1 ブートローダ	SPI1	有効	SPI1 設定 : - スレーブモード - 全二重 - 8 ビット MSB、最大 8 MHz の速度 - 極性 : CPOL ロー、CPHA ロー、NSS ハードウェア。
	SPI1_MOSI ピン	入力	PA7 ピン : スレーブ・データ入力ライン (プッシュプル、ブルダウンモードで使用)
	SPI1_MISO ピン	出力	PA6 ピン : スレーブ・データ出力ライン (プッシュプル、ブルダウンモードで使用)
	SPI1_SCK ピン	入力	PA5 ピン : スレーブ・クロック・ライン (プッシュプル、ブルダウンモードで使用)
	SPI1_NSS ピン	入力	PA4 ピン : スレーブ・チップ選択ピン (プッシュプル、ブルダウンモードで使用)

STM32F413 time

STM32F411xx デバイスのブートローダ

AN2606

表 67. システムメモリブートモードでの STM32F411xx の設定 (続き)

ブートローダ	機能/ペリフェラル	状態	コメント
SPI1 ブートローダ	SPI1	有効	SPI1 設定 : スレーブモード、全二重、8 ビット MSB、最大 8 MHz の速度、極性 : CPOL ロー、CPHA ロー、NSS ハードウェア。
	SPI1_MOSI ピン	入力	PA7 ピン : スレーブ・データ入力ライン (プッシュプル、ブルダウンモードで使用)
	SPI1_MISO ピン	出力	PA6 ピン : スレーブ・データ出力ライン (プッシュプル、ブルダウンモードで使用)
	SPI1_SCK ピン	入力	PA5 ピン : スレーブ・クロック・ライン (プッシュプル、ブルダウンモードで使用)
	SPI1_NSS ピン	入力	PA4 ピン : スレーブ・チップ選択ピン (プッシュプル、ブルダウンモードで使用)

STM32F411 hours

It seems to have the same problem as Momoko because it outputs the same.

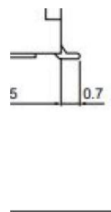
SR connector cable for writing and communication is damaged when plugged in or unplugged.

It tears twice a day



Cable Assembly Socket
[A04SR04SR30K152B](#)

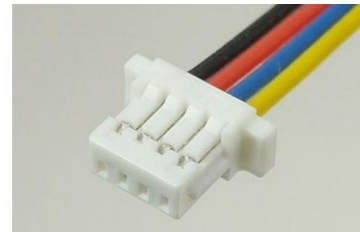
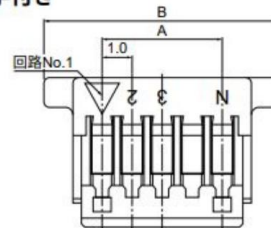
SRSH connector also available



コンタクト：銅合金・銅下地付すずめっき（リフロー処理）
ハウジング：ポリアミド・UL94V-0、ナチュラル（アイボリー）
補強金具：黄銅・銅下地付すずめっき（リフロー処理）

- RoHS2 対応品 本製品はラベルに(LF) (SN)を表示します。
- 上記製品はエンボステープ供給品です。
- 注 1) 金めっき仕様品についてはお問い合わせください。
- 2) SR コネクタ (圧接方式) と共用できます。
- 3) トップ型には吸着用のテープ付タイプも用意しております。
テープ付品についてはお問い合わせください。

取っ手付き



Should I use the SH connector

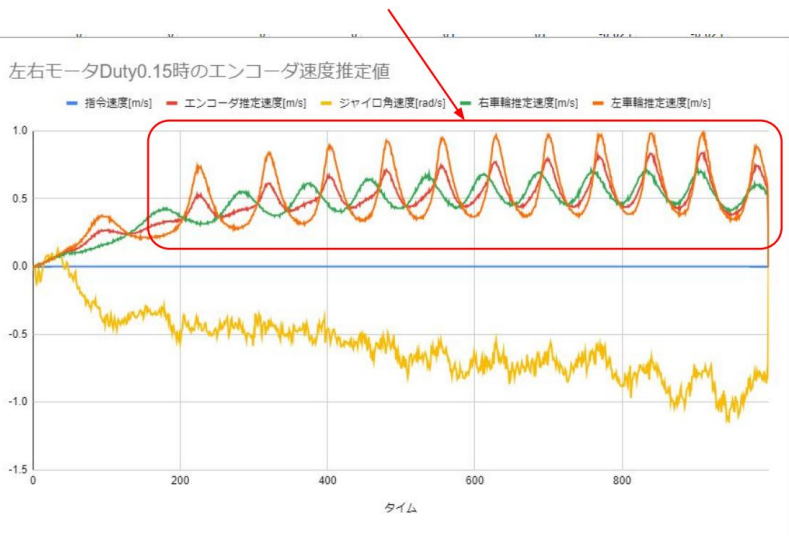
DG01032-0056-01 with a handle for easy insertion and

removal, and a cord ? (Should I ignore the black and red colors or swap the

Image source: [JST SH Connector Catalog](#), [Akizuki Electronics](#)

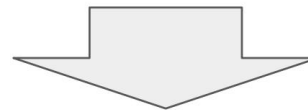
Encoder data is wavy

When I run it with a fixed duty cycle, the encoder data ripples...



Samarcobalt magnet (SS28), cylindrical, $\phi 1.9 \times 1$ (mm), radial surface magnetic flux density: 265 mT , magnetic encoder MA300 IC, magnet gap: ? (assumed 0.5 mm)

Warning! The current value of Bm (11.8mT) is below the minimum recommended value of 30mT.



Samarcobalt magnet (SS28), cylindrical, $\phi 3 \times 1$ mm, radial surface magnetic flux density: 218 mT , magnetic encoder MA730 IC, magnet gap: 0.5 mm

Warning! The current value of Bm (13.4mT) is below the minimum recommended value of 40mT.

Zirconia is also below the rated value. The minimum required magnetic flux density has increased due to the IC change, and v.1.1 seems to have deteriorated...

Reference: [MPS Online Magnet Calculator](#)

Encoder Acceleration Sensor Complementary Filter

It seems that vibrations are common when listening to what the superiors say. By using a velocity sensor in combination with filtering, the velocity can be estimated.

- Matsui: A method for measuring velocity by combining a velocity sensor and a rotary encoder https://matsui-mouse.blogspot.com/2014/03/blog-post_17.html • dangorogoro: Estimating velocity using a complementary filter for a micromouse (opposing two-wheeled vehicle) <https://blog.oino.li/posts/complementaryfilter/>

Programming level

Write using STM32CubeProgrammer Example of HEX
file size for writing: 277KB

The screenshot shows the STM32CubeProgrammer interface. The 'Memory & File editing' tab is active, displaying a memory map with columns for Address, 0, 4, 8, C, and ASCII. The 'Log' window at the bottom shows the following messages:

```

15:58:53 : Erasing internal memory sector 0
15:58:53 : Erasing memory corresponding to segment 1:
15:58:53 : Erasing internal memory sectors [2 4]
15:58:55 : Download in Progress:
15:59:05 : File download complete
15:59:05 : Time elapsed during download operation: 00:00:12.208
  
```

The progress bar at the bottom indicates 99% completion.

Log

```

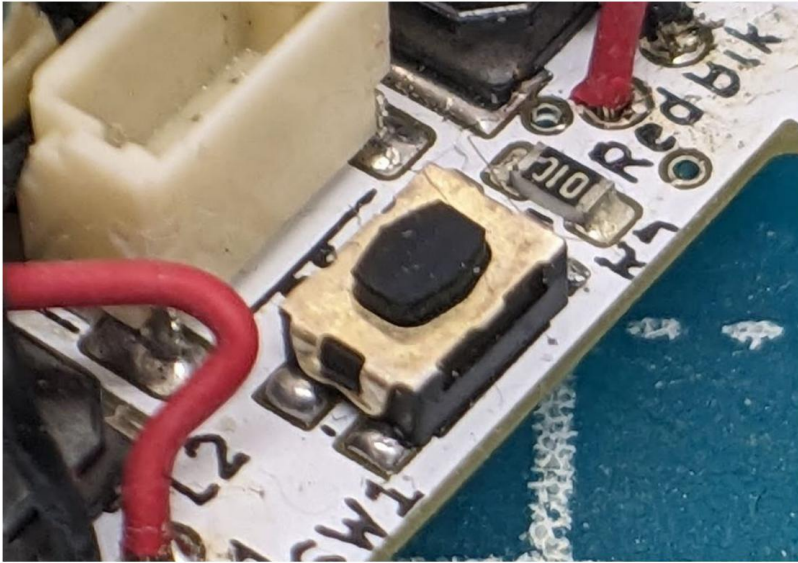
15:58:53 : Erasing internal memory sector 0
15:58:53 : Erasing memory corresponding to segment 1:
15:58:53 : Erasing internal memory sectors [2 4]
15:58:55 : Download in Progress:
15:59:05 : File download complete
15:59:05 : Time elapsed during download operation: 00:00:12.208
  
```

12.208 seconds

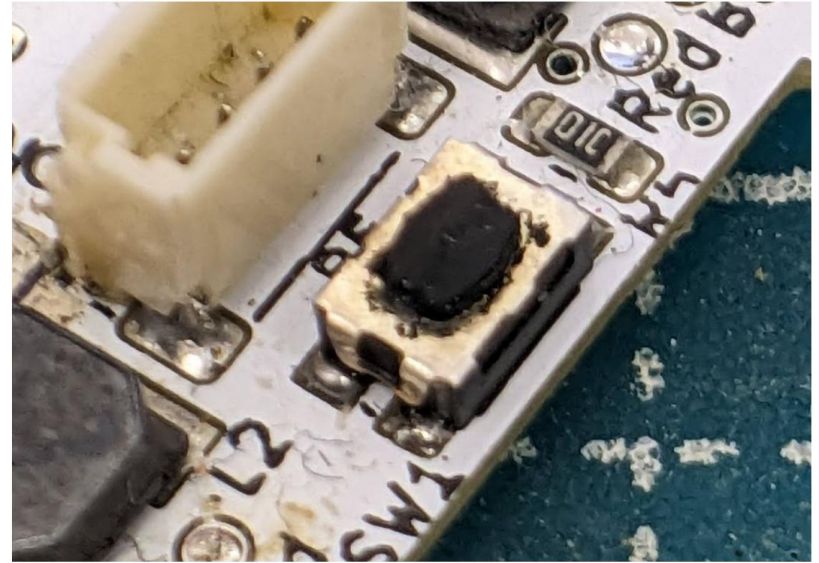
When adjusting
parameters, you have to
write frequently, which takes a lot of time.

Press the push button too frequently and it will get crushed.

During program development, the frequency of writing is high, so the push button is pressed frequently, which causes the button to become jammed and become unresponsive.



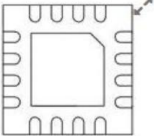
New Button



Poorly responsive buttons

Magnetic encoder: MA300 ÷ MA730

ABZ incremental 10-bit
output encoder

画像はあくまで参考です
製品仕様をご覧ください

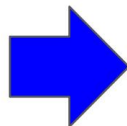
共有

Mouser 番号 :	946-MA300GQ-Z
メーカー 番号 :	MA300GQ-Z
メーカー :	Monolithic Power Systems (MPS)
お客様番号:	お客様番号
詳細 :	基板取付型ホール効果/磁気センサー Sensor for 3-Phase Brushless Motor
ライフサイクル:	LC 製造中止
データシート:	MA300GQ-Z データシート (PDF)
ECADモデル:	  PCB シンボル、フットプリント、および 3D モデル

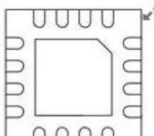
無料のライブラリローダーをダウンロードし、お使いのECADツール用にこのファイルを変換してください。[ECADモデルの詳細について](#)

Discontinuation

correspondence



ABZ incremental 12-bit
output encoder

画像はあくまで参考です
製品仕様をご覧ください

共有

Mouser 番号 :	946-MA730GQ-Z
メーカー 番号 :	MA730GQ-Z
メーカー :	Monolithic Power Systems (MPS)
お客様番号:	お客様番号
詳細 :	基板取付型ホール効果/磁気センサー 14-bit, digital, contactless angle sensor with ABZ incremental and PWM output
データシート:	MA730GQ-Z データシート (PDF)
ECADモデル:	  PCB シンボル、フットプリント、および 3D モデル

無料のライブラリローダーをダウンロードし、お使いのECADツール用にこのファイルを変換してください。[ECADモデルの詳細について](#)

It is better than the equivalent product MA302

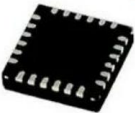
Higher resolution would be better.

Image source:

[Mouser site MA300GQ-Z](#)

[Mouser Site MA730GC-Z](#)

IMU: ICM-20648 ÷ ICM-42688-P



もう一枚の画像

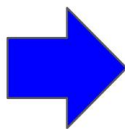
画像はあくまで参考です
製品仕様をご覧ください

共有

ICM-20648	
Mouser 番号 :	410-ICM-20648
メーカー 番号 :	ICM-20648
メーカー :	TDK InvenSense
お客様番号:	お客様番号
詳細 :	IMU - 慣性測定ユニット DMP-Enabled 6-Axis Integrated Solution for Wrist-Worn Wearable
ライフサイクル:	LC 製造中止: メーカーにてモデル廃止予定のため、製造中止となります。
データシート:	ICM-20648 データシート (PDF)
ECADモデル:	 PCBシンボル、フットプリント、3Dモデル
無料のライブラリローダーをダウンロードし、お使いのECADツール用にこのファイルを変換してください。ECADモデルの詳細について	

Discontinuation

correspondence





画像はあくまで参考です
製品仕様をご覧ください

共有

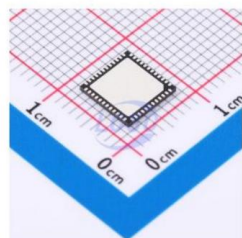
ICM-42688-P	
Mouser 番号 :	410-ICM-42688-P
メーカー 番号 :	ICM-42688-P
メーカー :	TDK InvenSense
お客様番号:	お客様番号
詳細 :	IMU - 慣性測定ユニット 6 Axis IMU Motion Tracking
データシート:	ICM-42688-P データシート (PDF)
ECADモデル:	 PCBシンボル、フットプリント、3Dモデル
無料のライブラリローダーをダウンロードし、お使いのECADツール用にこのファイルを変換してください。ECADモデルの詳細について	
詳細を表示	TDK InvenSense ICM-42688-Pの詳細を読む

There is plenty of stock on the market,
so it's best to change it.

Image source:

[Mouser site ICM-20648](#)
[Mouser site ICM-42688-P](#)

Microcontroller: STM32F413 ü STM32F411



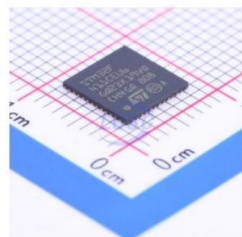
* Images are for reference only

STM32F413CHU6

Manufacturer	STMicroelectronics
MFR. Part #	STM32F413CHU6
JLCPCB Part #	C1337761
Package	UFQFPN-48(7x7)
Description	1.5MB 1.7V~3.6V ARM Cortex-M4 320KB 100MHz FLASH 36 UFQFPN-48(7x7) Microcontroller Units (MCUs/MPUs/SOCs) ROHS
Datasheet	Download
Source	JLCPCB
Assembly Type	SMT Assembly
CAD Model	PCB Footprint or Symbol

Note: The purchased components are stored in your JLCPCB parts library for future PCBA orders only, and cannot be shipped separately.

Assembly Tips Specifications PCB footprint/Symbol Documents



* Images are for reference only

STM32F411CEU6

Manufacturer	STMicroelectronics
MFR. Part #	STM32F411CEU6
JLCPCB Part #	C60420
Package	UFQFPN-48(7x7)
Description	512KB 1.7V~3.6V ARM Cortex-M4 128KB 100MHz FLASH 36 UFQFPN-48(7x7) Microcontroller Units (MCUs/MPUs/SOCs) ROHS
Datasheet	Download
Source	JLCPCB
Assembly Type	SMT Assembly
CAD Model	PCB Footprint or Symbol

Note: The purchased components are stored in your JLCPCB parts library for future PCBA orders only, and cannot be shipped separately.

In Stock: 0

Minimum: 1 Full Reel: 260

Estimated unit price: \$13.5570

Total: \$13.557 [Parts Calculator](#)

[Add to My Part Lib for Assembly](#)

[Add To List](#)

Available Order Qty: 0

In-stock Item Pricing

Insufficient stock, pre-order mode turns on

STM32F413 out of stock. Pin compatible STM32F411 change.
(Zirconia circuit diagram also included)

FLASH: 1.5MB ü 512KB RAM:
320KB ü 128KB

In Stock: 3607

Minimum: 1 Full Reel: 260

Total: \$2.9745 [Parts Calculator](#)

[Add to My Part Lib for Assembly](#)

[Add To List](#)

Available Order Qty: 3292

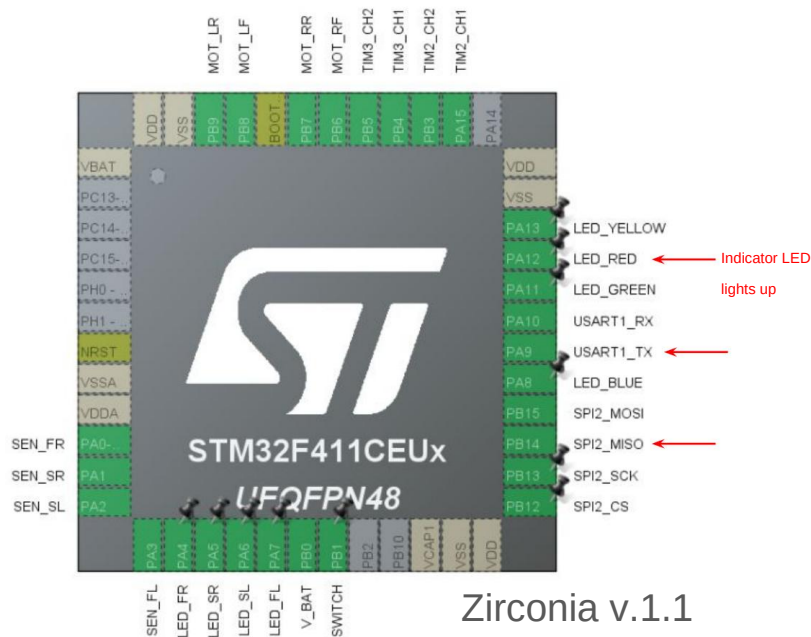
In-stock Item Pricing

Image source:

[JLCPCB STM32F413CHU6](#)

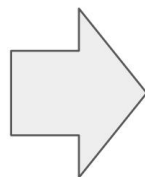
[JLCPCB STM32F411CEU6](#)

Pin layout changed to resolve issues in boot mode



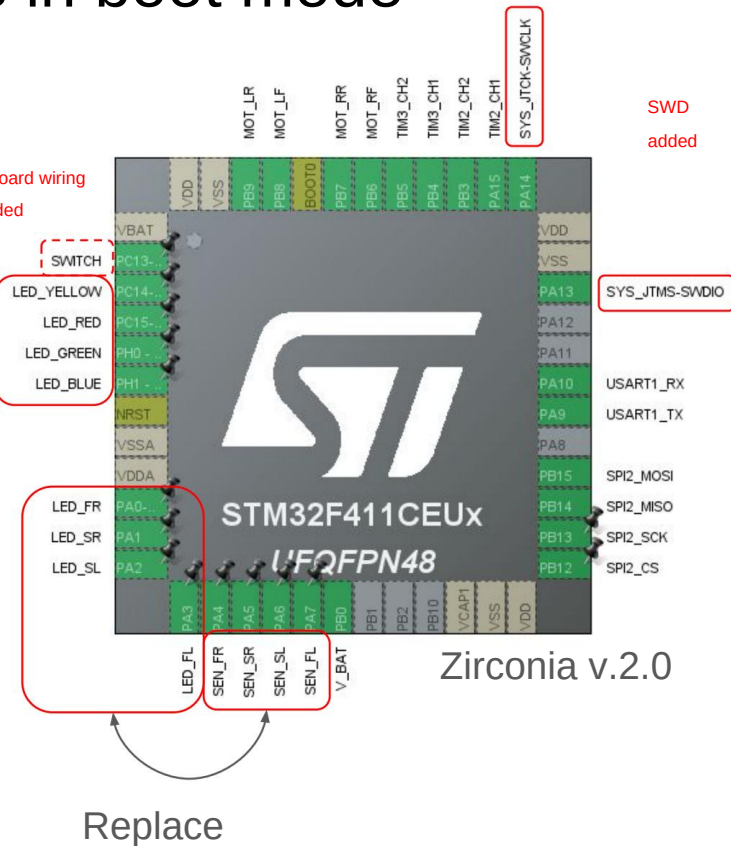
The wall sensor LED is ON
and a large current flows.

Pin that goes high in boot mode



move

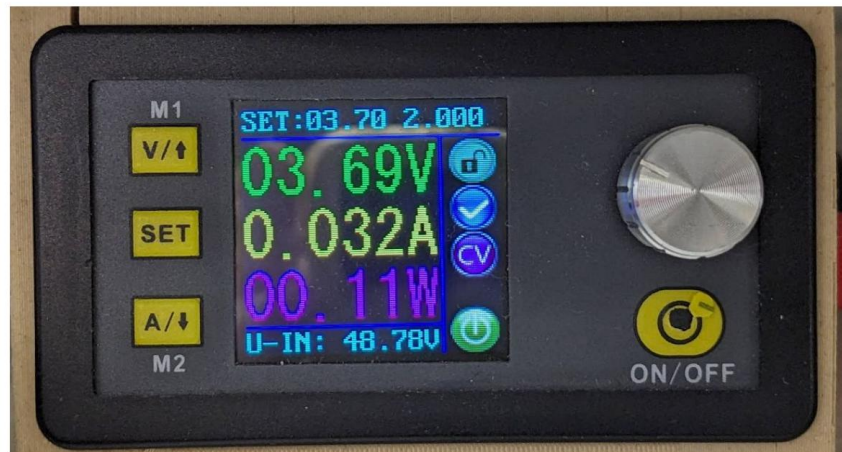
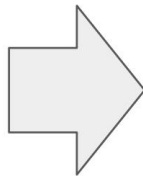
Move board wiring
as needed



Eliminates current flow during boot mode (semi-short state)

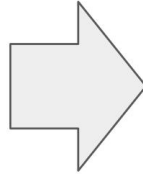
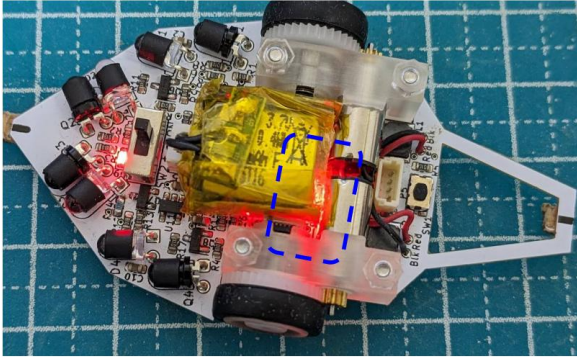


Zirconia
v.1.1 Boot mode: 0.225A

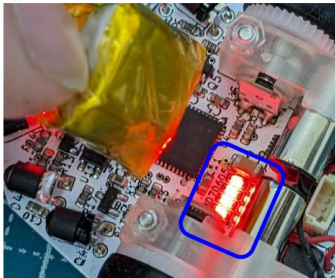


Zirconia v.2.0
Boot mode: 0.032A

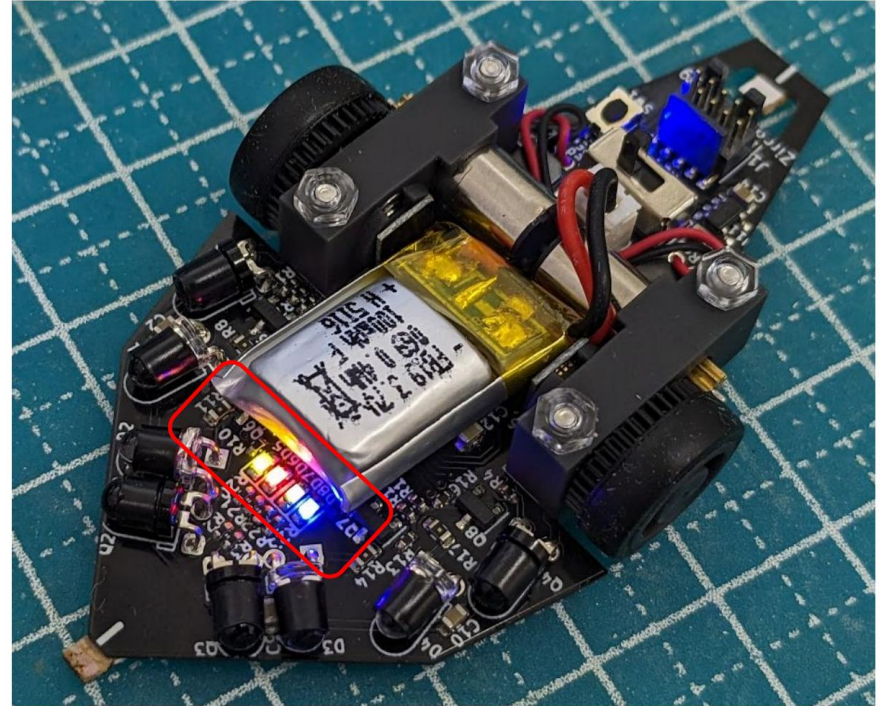
Change the LED position so that the indicator LED is hidden by the battery



Move the
battery
up and look inside



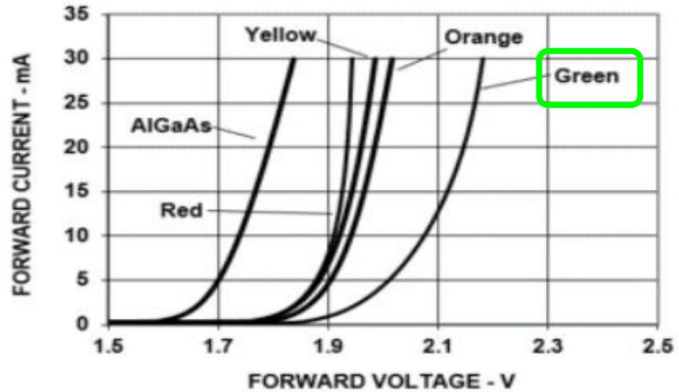
Zirconia v.1.1



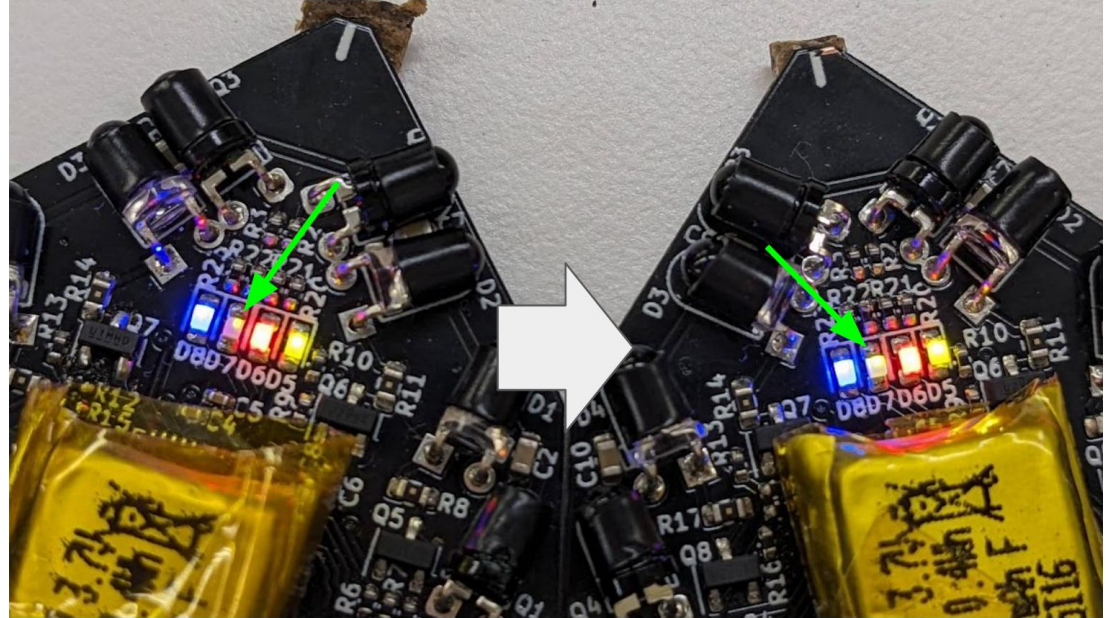
Zirconia v.2.0

Increased brightness of green LED indicator

Figure 2: Forward Current vs. Forward Voltage



Green LED model number: HSMG-C190 High Vf compared to other colors
Reduce current limiting resistor

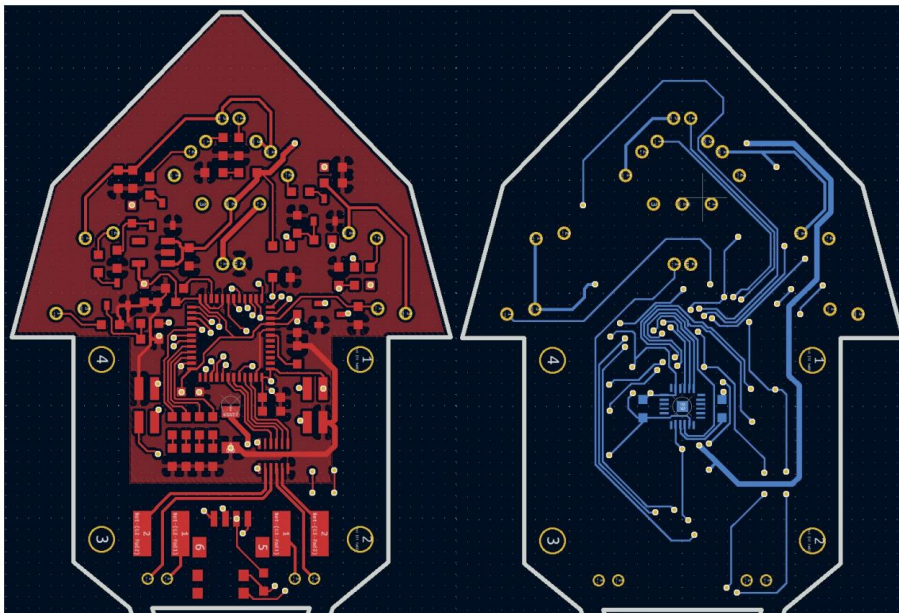


Before change (1k Ω) After change (470 Ω)

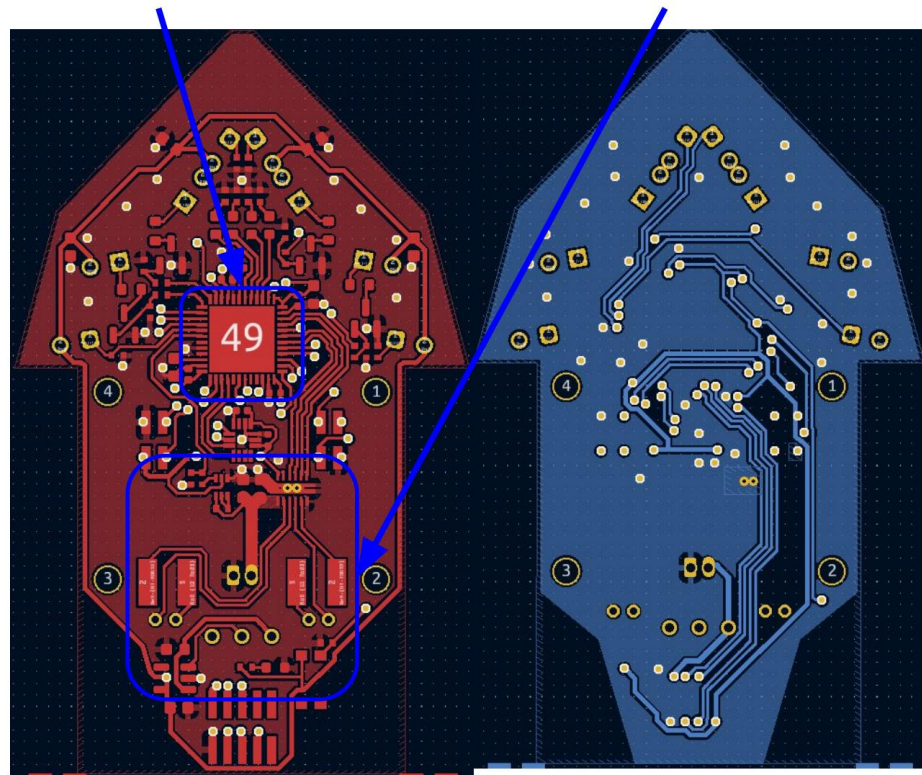
Image source: [Broadcom datasheet](#)

PCB wiring arrangement

Remove vias from the pad under the microcontroller. Lower the power supply system.

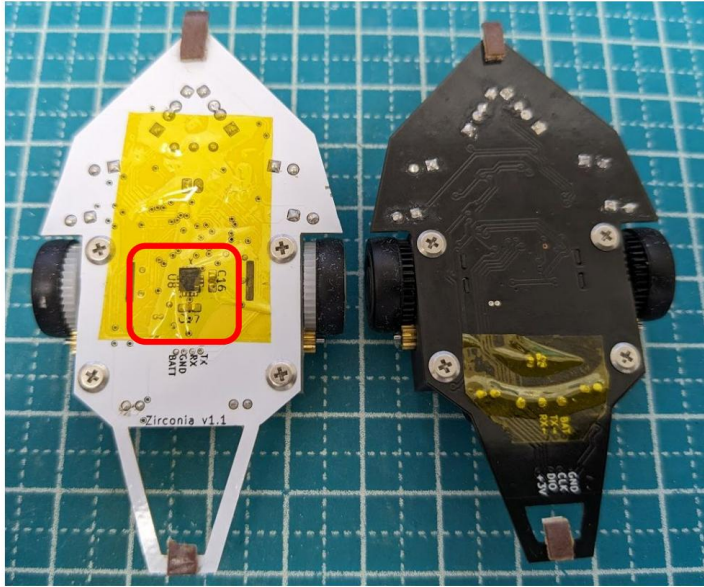


Zirconia v.1.0



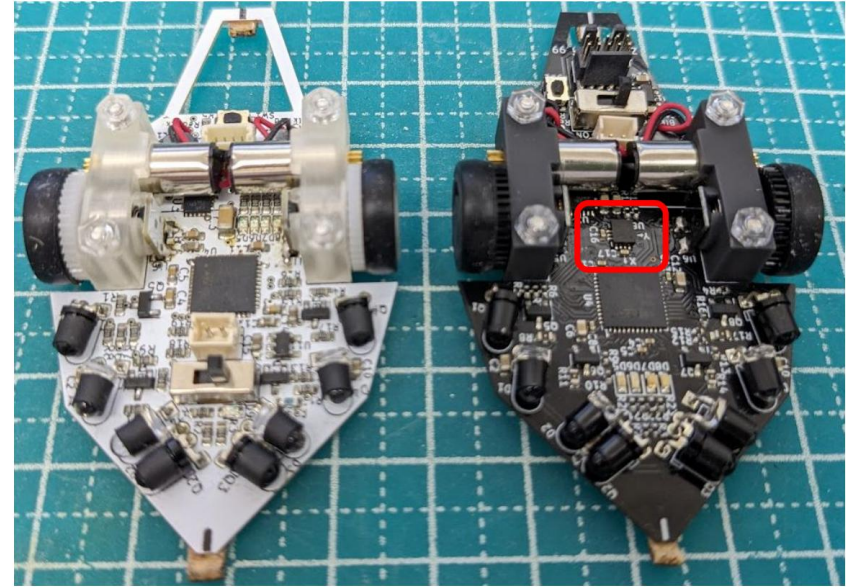
Zirconia v.2.0

All mounted components on the surface (IMU circuit components)



Zirconia v.1.1

Zirconia v.2.0

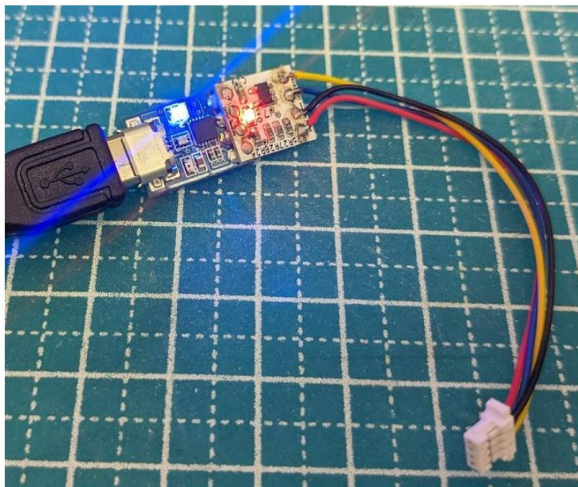


Zirconia v.1.1

Zirconia v.2.0

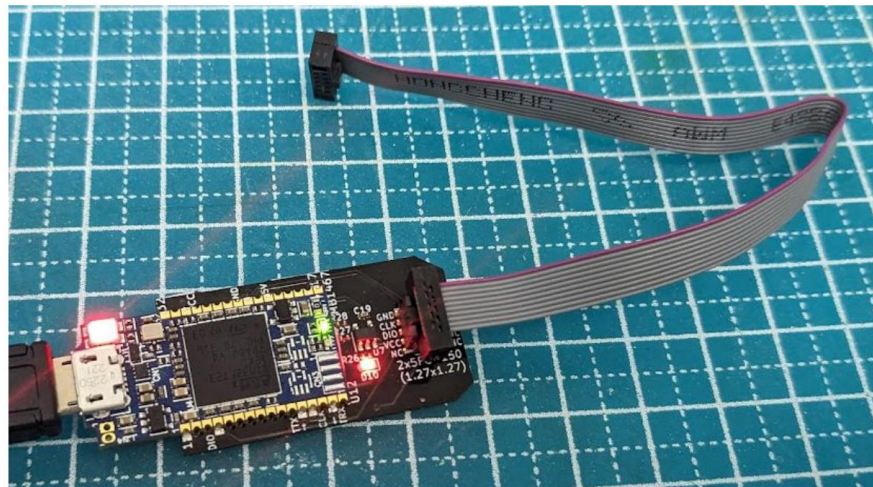
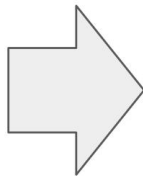
Back side No mounted components

Added ST-LINK



USB-UART
conversion + Li-ion battery charging

Zirconia v.1.1



SWD (Serial Wire Debug)
+USB-UART
conversion +Lithium-ion battery charging

Zirconia v.2.0

Improved programmability via SWD

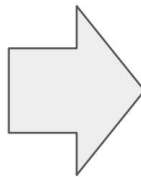
Example of HEX file size for writing: 227KB

Log

```
15:58:53 : Erasing internal memory sector 0  
15:58:53 : Erasing memory corresponding to segment 1:  
15:58:53 : Erasing internal memory sectors [2 4]  
15:58:55 : Download in Progress:  
15:59:05 : File download complete  
15:59:05 : Time elapsed during download operation: 00:00:12.208
```

12.208 seconds

Zirconia v.1.1



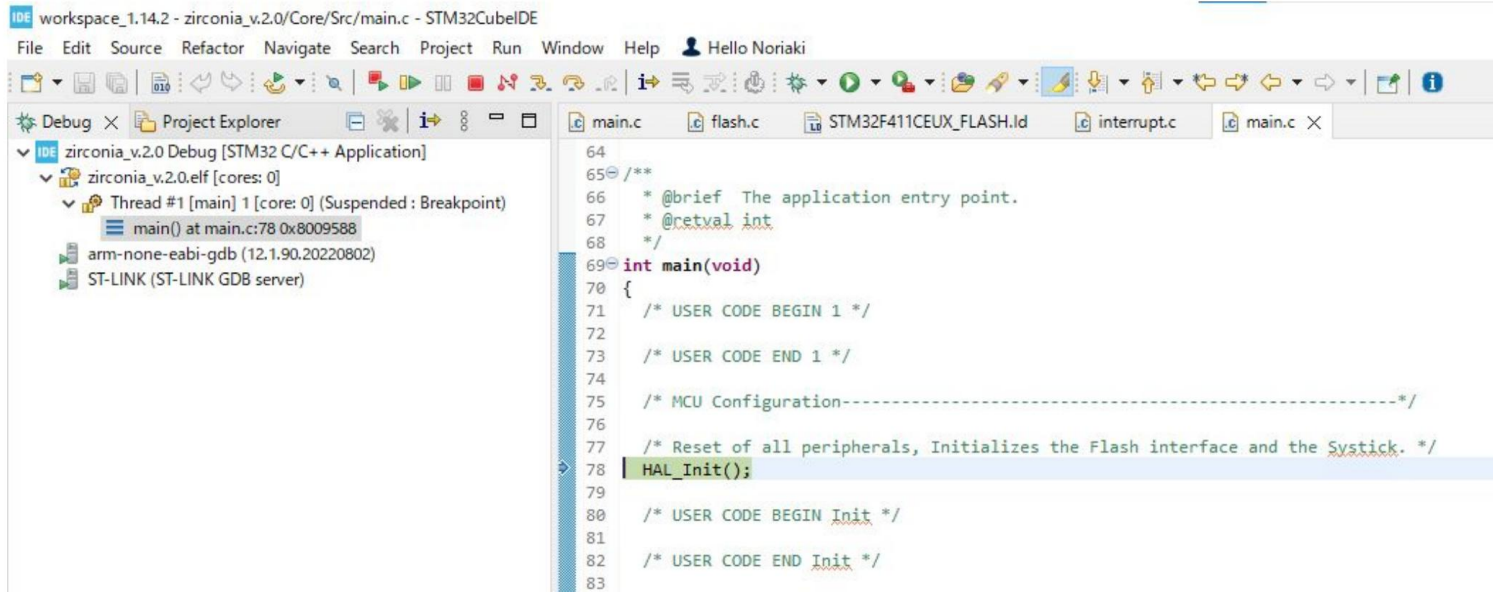
Log

```
16:00:04 : Erasing internal memory sector 0  
16:00:04 : Erasing memory corresponding to segment 1:  
16:00:04 : Erasing internal memory sectors [2 4]  
16:00:05 : Download in Progress:  
16:00:06 : File download complete  
16:00:06 : Time elapsed during download operation: 00:00:01.778
```

1.778 seconds

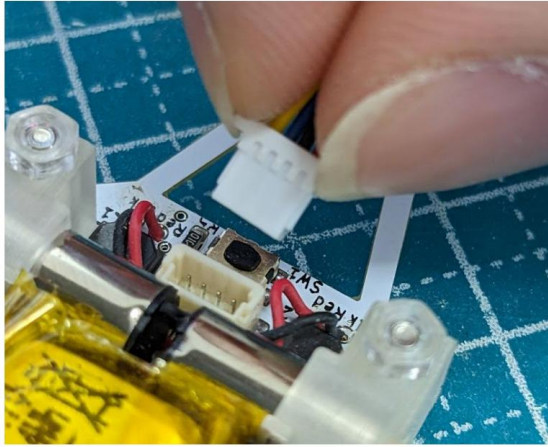
Zirconia v.2.0

To enable debugging



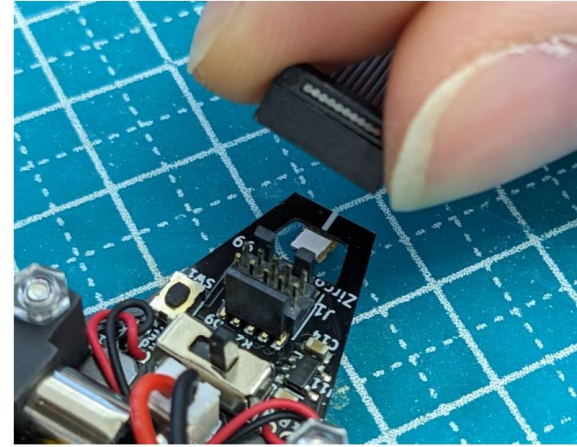
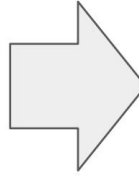
STM32CubeIDE debug screen

Change the communication connector type



Zirconia v.1.1

SR(SH) connector claw removal

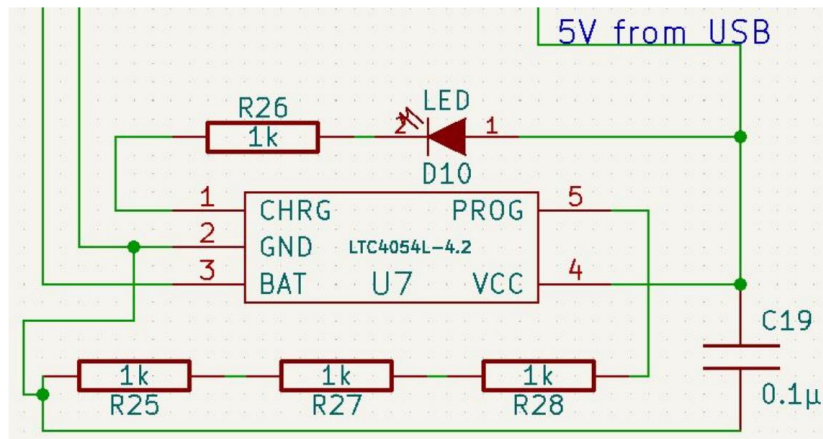


Zirconia v.2.0

IDC Connector

Easy to insert and remove

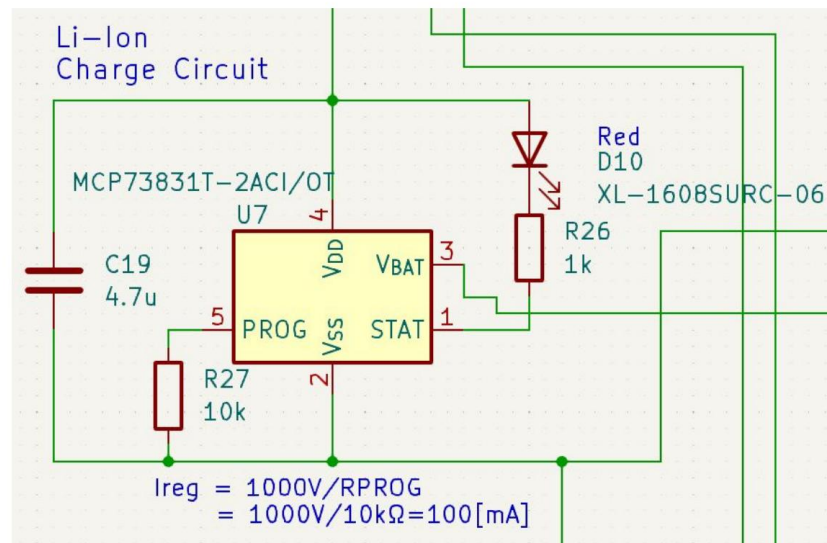
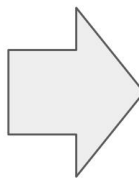
Battery charge level increases slightly



50mA charging

(assuming a 50mAh battery)

Zirconia v.1.1



100mA charging

(assuming 100mAh battery)

(low-cost IC change)

Zirconia v.2.0

Magnet change position change

ネオジム磁石(N40)、円柱型、Φ3x2(mm)、径方向



直径D = 3 (mm)
高さH = 2 (mm)

製品詳細	
種類	ネオジム磁石
形状	円柱型
寸法	Φ3x2
寸法公差	D(0/-0.1) x H(0/-0.1) (mm)
表面処理	Niメッキ(NiCuNi)
表面磁束密度	3,970 G / 397 mT
吸着力	0.14 kgf
材質記号	N40
限界使用温度	140℃

購入数量	1個	単価再計算
※購入数量によって、単価が変わります。		
本体価格	49円/個 (税込価格 53.9円)	
合計金額	49円 (税込 53.9円)	
在庫状況	在庫あり	
発送予定日	当日出荷可能 (但し、17時までのご注文に限る。詳細>>)	
ネット特典	ポイント 1% 付与 、ネット注文割引 5%	

買物カゴに入れる

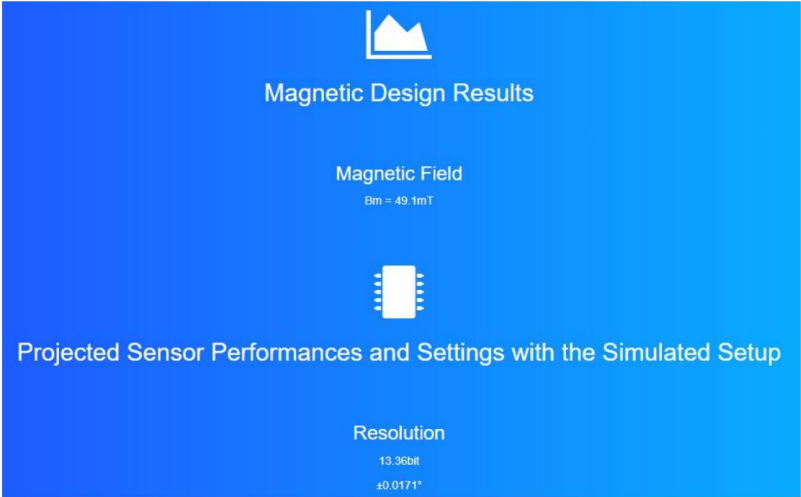
- ご注文の前にお読み下さい。(購入ガイド)
- 返品に関するお知らせ
- この商品についてお問い合わせ
- この商品の見積り依頼

※表面磁束密度と吸着力の値は参考値です。
※測定方法はこちらをご参照下さい。
※単価は原材料価格により変動する場合がございます。前回、ご参照された単価と異なる場合がございます。予めご了承ください。

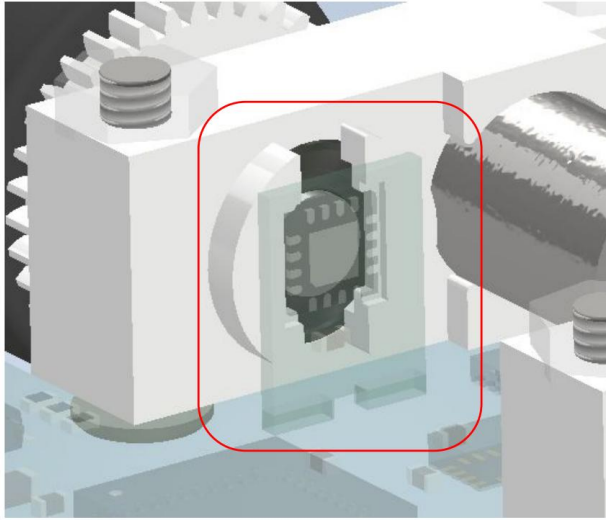
Inputs

Parameters used to perform the magnet simulation

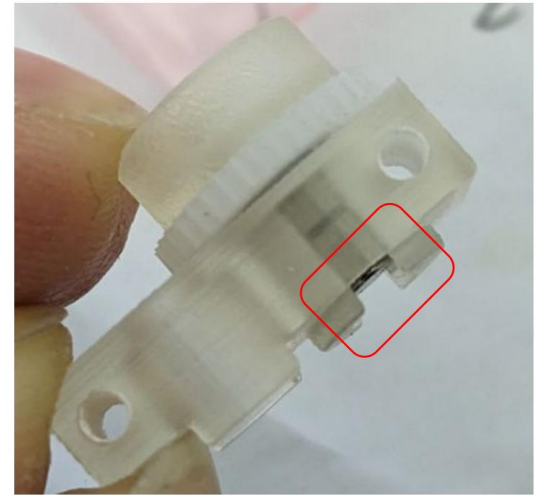
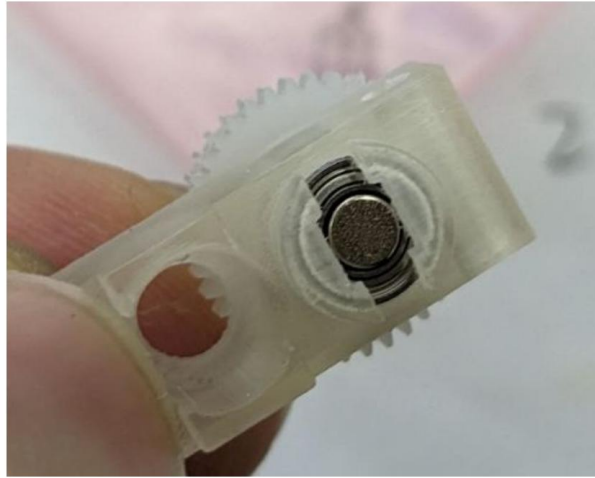
Symbol	Value	Description
Sensor Location	End-of-Shaft	
Magnet Type	Cylinder	
Brem	0.397T	Magnet Remanence
H	2.0mm	Magnet Height
OD	3.0mm	Magnet Outer Diameter
r	0mm	Radius Between Magnet Center and Sensor Sensitive Point
z	1.6mm (0.1mm air gap)	Z offset Between Magnet Center and Sensor Sensitive Point
Sensor Part Number	MA730	
Sensor Size	3 x 3 x 1 [mm]	Length x Width x Height (LxWxH)
Sensor Filter Cutoff Frequency	23Hz	



Encoder board positioning jig and motor mount

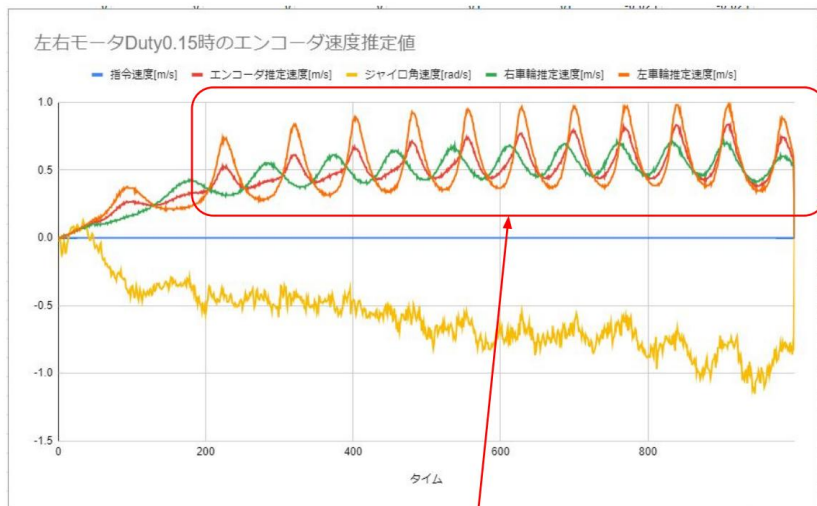


The sensor IC mounted on the encoder board fits into the motor mount.



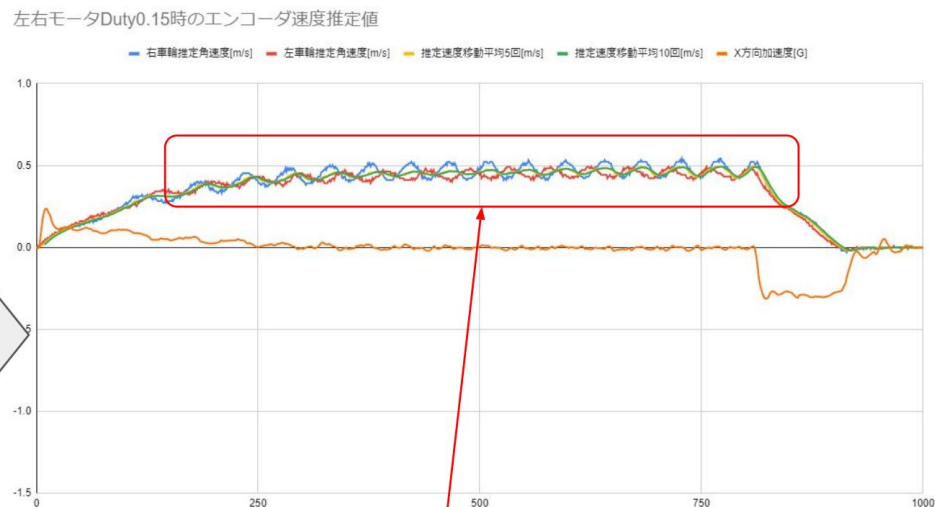
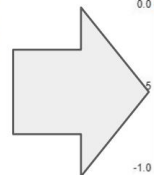
Remove the magnet
and move the sensor IC closer.

Effect: Constant Duty Ratio Estimation



Zirconia v.1.1

When I run it with a fixed duty cycle,
the encoder data ripples...



Zirconia v.2.0

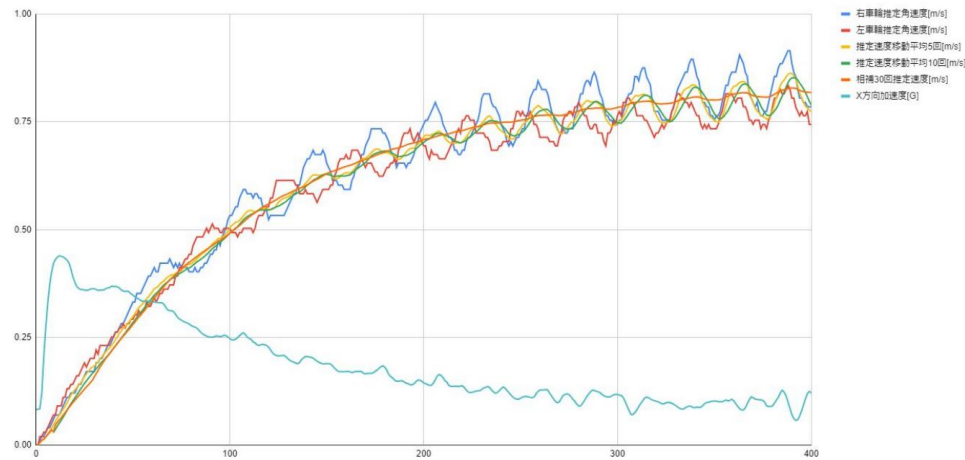
It has been greatly improved

Magnetic encoder IC upgraded from MA302 to high resolution MA730

Left and right motor duty ratio: 0.2, encoder measurement accuracy (measured at 1 kHz)



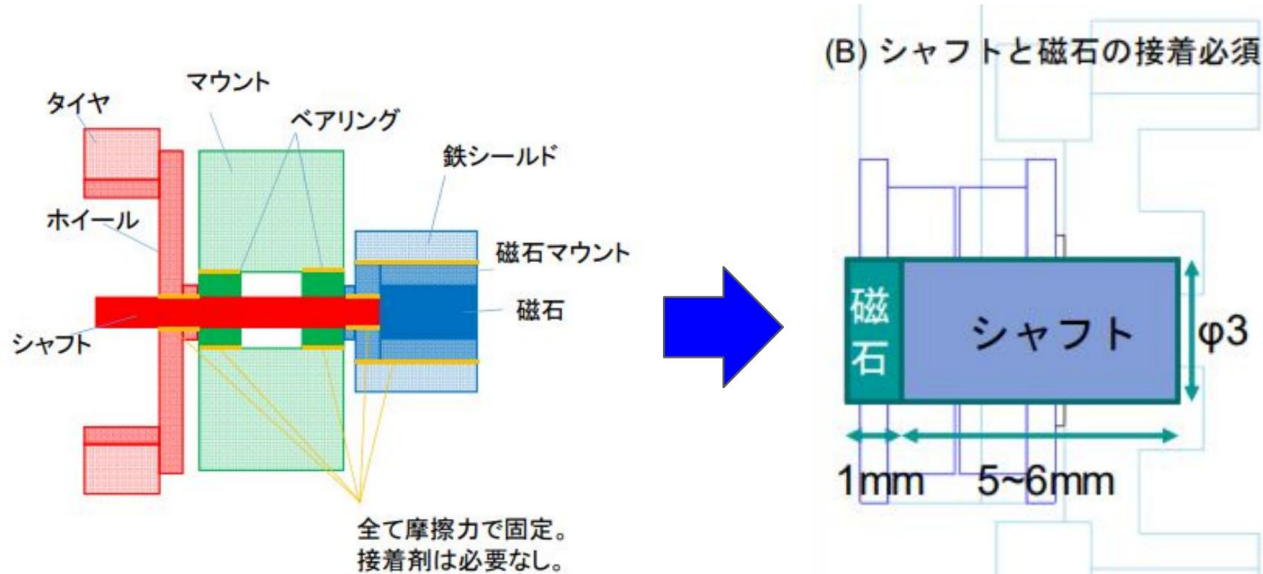
For MA302



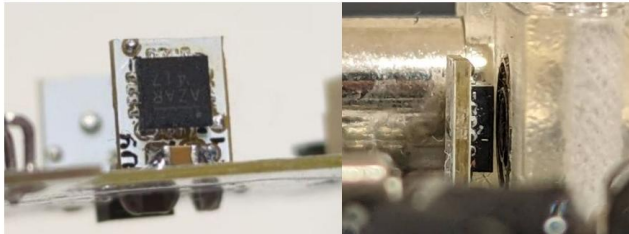
For MA730

Reduced stuttering

Undercarriage: Bearing holder system



Bearing Magnet



Opis-san

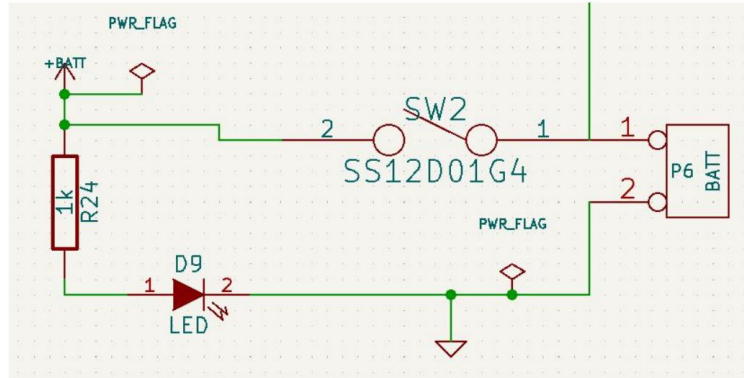
Image source:

[Zirconia Production Manual](#)

Micromouse Undercarriage

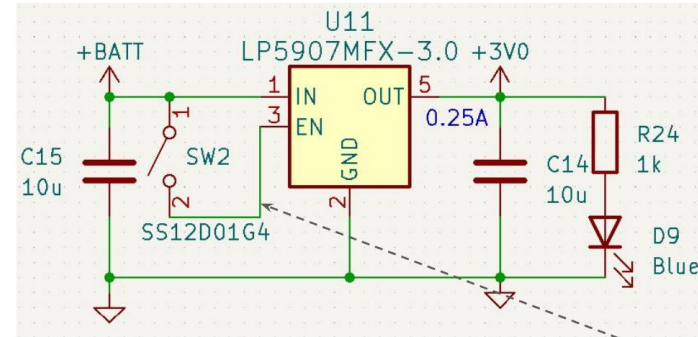
It follows the "Micromouse Undercarriage: Introducing the Simple Undercarriage as Envisioned by a Software Maker."

Changes due to board wiring

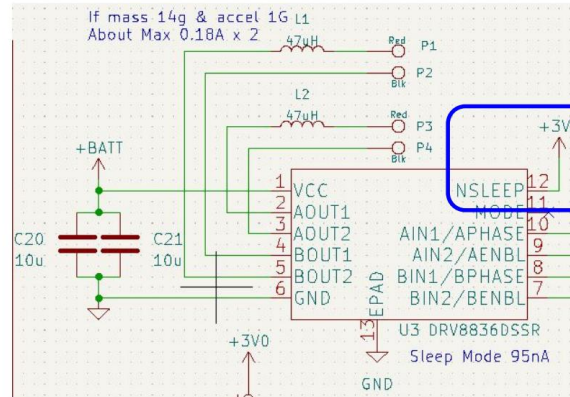


Zirconia v.1.0

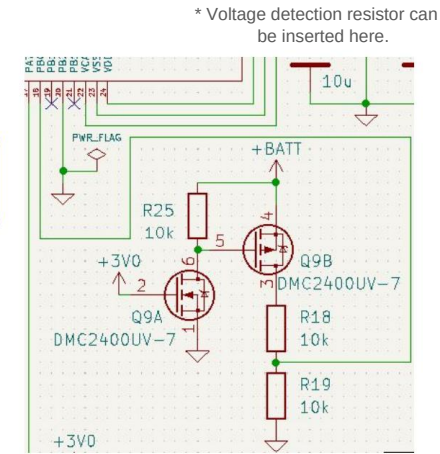
Other, assuming implementation
request, change from 1608
size product to 1005 size product



Regulator Enable pin Power ON/OFF



Sleep mode 95nA consumption



Battery voltage
detection FET cuts consumption.

Zirconia v.2.0

JLCPCB assembly request method

JLCPBC implementation request 1 data

workspace / nnakagawa / Zirconia_Sample

production

Here's where you'll find this repository's source files. To give your users an idea of what they'll find here, [add a description to your repository](#)



v.2.0

Files

Filter files



zirconia_sample / hardware / production

Name	Size	Last commit	Message
..			
backups		2024-06-07	
Zirconia_v.2.0.zip	233.07 KB	2024-06-07	update
bom.csv	1.54 KB	2024-06-07	update
designators.csv	501 B	2024-06-07	update
netlist.ipc	31.55 KB	2024-06-07	update
positions.csv	2.25 KB	2024-06-07	update

Implementation request required data 3

Zirconia_v.2.0.zip
Gerber data

bom.csv
Parts data

positions.csv
Part placement data

JLCPBC implementation request 2 Gerber data upload

Standard PCB/PCBA

Limited Time Offer
Advanced PCB/PCBA

SMT-Stencil

3D Printing/CNC



[← Back to Upload File](#)Detected 2 layer board of 91.3x39.38mm(3.59x1.55 inches).[Gerber Viewer](#)

Base Material

FR-4

Flex

Aluminum

Copper Core

Rogers

PTFE Teflon

Layers

1

2

4

High Precision PCB

6

8

10

12

14

16

18

20

Dimensions

39.38

*

91.3

mm

PCB Qty

5

Product Type

Industrial/Consumer electronics

Aerospace

Medical

Zirconia_v.2.0.zip
Gerber data
upload screen

JLCPBC Mounting Request 3: Selecting the board thickness

3 types of boards:
main body, encoder board,
writing & charging board

PCB Specifications

Different Design	☐ 1	☐ 2	☒ 3	☐ 4	☐		
Delivery Format	☐ Single PCB	☒ Panel by Customer	☐ Panel by JLCPBC				
PCB Thickness	☐ 0.4	☒ 0.6	☐ 0.8	☐ 1.0	☐ 1.2	☐ 1.6	☐ 2.0
PCB Color	☒ Green	☐ Purple	☐ Red	☐ Yellow	☐ Blue	☐ White	☐ Black
Silkscreen	☒ White						
Surface Finish	☐ HASL(with lead)	☒ LeadFree HASL	☐ ENIG				

Select
0.6mm

Select your color preference

Delivery time and price will increase

The solder to be placed on the pad is leaded, lead-free, or gold-plated.

JLCPBC Implementation Request 4 Select the number of implementations

PCB Assembly

COUPON Free Assembly for your PCB order

ON

PCB Assembly

COUPON Free Assembly for your PCB order

Assemble top side

Assemble bottom side

PCBA Type

Economic Standard What's the difference?

Assembly Side

Top Side Bottom Side Both Sides

PCBA Qty

5 Custom Qty Qty from 2 to 5

Edge Rails/Fiducials

Added by JLCPCB Added by Customer

The board size is modified to be 71.38mm*91.3mm due to adding two 5mm edge rails on the shorter sides. [Learn More>](#)

Confirm Parts Placement

No Yes

Special changes are OK as shown in the image

Enter the desired number of cards. Default: 5 cards.

JLCPBC Implementation Request 5 Custom Settings

Advanced Options

Photo Confirmation ?

No

Board Cleaning ?

No

Conformal Coating ?

No

Bake Components ?

No

Packaging ?

Antistatic bubble film

Depanel boards & edge rail before delivery ?

No

Solder Paste ?

Sn96.5%, Ag3.0%, C

Others ?

No

Add paste for unpopulated pad & step stencil opening ?

No

Remark

☒ I agree to [the Terms and Conditions of JLCPCB assembly Service](#)

Confirm

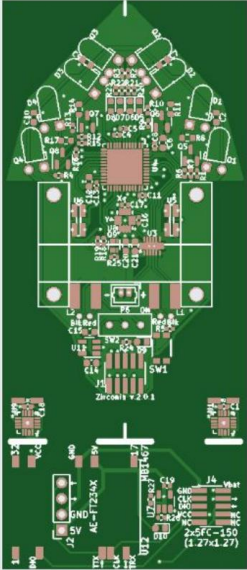
Special changes

Check the board

[PCB](#) [Bill of Materials](#) [Component Placements](#) [Quote & Order](#)

Automatically saved, last updated on 22 June, 15:43

Top Bottom



[← Change PCB specifications](#)
PCBA Type: Standard
Assembly Side: Top Side
PCBA Qty: 5

NEXT

Next

Upload part data and location data

The screenshot shows a web interface for a PCB design tool. At the top, the project name 'Zirconia_v2.0' is displayed. Below it, a navigation bar includes 'PCB', 'Bill of Materials' (which is the active tab), 'Component Placements', and 'Quote & Order'. A status message on the right says 'Automatically saved, last updated on 22 June, 15:46'. The main content area has two large light blue boxes. The left box contains a blue button labeled 'Add BOM File' with an upload icon, and text below it stating 'Only accept XLS,XLSX,CSV. [View Sample BOM](#)'. A callout box points to this section, containing the text 'bom.csv' and 'Parts data'. The right box contains a blue button labeled 'Add CPL File' with an upload icon, and text below it stating 'Pick&Place File, Only accept XLS,XLSX,CSV. [View Sample CPL](#)'. A callout box points to this section, containing the text 'positions.csv' and 'Part placement data'. At the bottom left, there is a link 'Not sure where to start? Check our [SMT FAQs](#)'. At the bottom right, a button labeled 'Process BOM & CPL' is partially visible.

Zirconia_v2.0

PCB Bill of Materials Component Placements Quote & Order

Automatically saved, last updated on 22 June, 15:46

Add BOM File

Only accept XLS,XLSX,CSV. [View Sample BOM](#)

Not sure where to start? Check our [SMT FAQs](#)

bom.csv
Parts data

Add CPL File

Pick&Place File, Only accept XLS,XLSX,CSV. [View Sample CPL](#)

Process BOM & CPL

positions.csv
Part placement data

Parts data confirmation screen 1

Drone_k28 - J
PCB Bill of Materials Component Placement Guide & Order Automatically saved, last updated on 22 June, 15:48

Top Side									
Total: 36 parts detected 26 parts to be confirmed 10 parts not selected									
Uploaded BOM Data				Review Matched Parts					
Top Designer	Comment	Footprint	Matched Part Detail	Qty	Source	Lib Type	Total Cost	Select	
C15H1,C15H1E...	0.1u	0402	CL35H15K050NNNG C1525	30	JLCPCB	Basic	¥5,980.00	☐	☐
C16,C162,C164,C1...	10u	0603	CL35H15K050NNNG C16700	60	JLCPCB	Basic	¥49,547.91	☐	☐
C17	10u	0402	CL35H15K050NNNG C17100	20	JLCPCB	Basic	¥3,834.41	☐	☐
C18	4.7u	0402	CL35H15K050NNNG C18700	20	JLCPCB	Basic	¥16,618.00	☐	☐
D5	XL-1608LVC...	0603	XL-1608LVC-06 C060802	20	JLCPCB	Extended	¥17,899.00	☐	☐
D7	XL-1608LVC...	0603	XL-1608LVC-06 C060803	20	JLCPCB	Extended	¥17,874.44	☐	☐
D8,D9	XL-1608LVC...	0603	XL-1608LVC-06 C060807	20	JLCPCB	Extended	¥10,842.41	☐	☐
D10,D6	XL-1608LVC...	0603	XL-1608LVC-06 C060789	20	JLCPCB	Extended	¥5,580.00	☐	☐
J1,J4	BD125-15A...	PowerSoc_2x6	BD125-15A-01 C0000000	10	JLCPCB	Extended	¥108,801.93	☐	☐
L1,L2	47uH	Inductor	47uH-47uH-01 C0000000	10	JLCPCB	Extended	¥389,519.19	☐	☐
Q1,Q5,Q7,Q8	RLM4E344	ICOT-23	RLM4E344 C0000000	24	JLCPCB	Extended	¥154,916.01	☐	☐
Q9	DMC2400V-7	ICOT-563	DMC2400V-7 C177025	9	JLCPCB	Extended	¥89,862.41	☐	☐
R1,R12,R15,R2...	1k	0402	9402WGP1001TCE C17702	75	JLCPCB	Basic	¥5,913.13	☐	☐
R10,R13,R16,R7...	100k	0402	9402WGP1001TCE C0741	30	JLCPCB	Basic	¥5,366.00	☐	☐
R11,R14,R17,R8	10	0603	9402WGP1001TCE C2269	30	JLCPCB	Basic	¥4,317.00	☐	☐
R16,R19,R20,R2...	10k	0402	9402WGP1001TCE C0704	25	JLCPCB	Basic	¥1,801.11	☐	☐
R22	510	0402	9402WGP1001TCE C0702	20	JLCPCB	Basic	¥1,907.71	☐	☐
S01	XMR2-110G...	XMR2-110G_LF3	XMR2-110G_LF3 C0000000	5	JLCPCB	Extended	¥323,526.02	☐	☐
U1	DRV8830DSR	DRV8830DSR	DRV8830DSR C0000000	5	JLCPCB	Extended	¥804,202.22	☐	☐
U4	STM32F411...	STM32F411...	STM32F411... C0000000	5	JLCPCB	Extended	¥104,022.02	☐	☐
U7	MCPT2831T...	MCPT2831T...	MCPT2831T... C0000000	5	JLCPCB	Extended	¥541,204.01	☐	☐
U8	ICM-42688-P...	ICM-42688-P...	ICM-42688-P... C0000000	5	JLCPCB	Extended	¥184,154.13	☐	☐
U11	LP5557MFA...	LP5557MFA...	LP5557MFA... C0000000	5	JLCPCB	Extended	¥281,362.23	☐	☐

For below not matched parts, try our Manual Parts Selection >

D1,D3	OSI5FU3A11C	OSI5FU3A11C	No Part Selected	☐	☐
D2,D4	OSI5FU3A11C	OSI5FU3A11C...	No Part Selected	☐	☐
J2	Conn_01x04	PinSocket_1x0...	No Part Selected	☐	☐
P1,P3	Red	motor_pin	No Part Selected	☐	☐
P2,P4	Blk	motor_pin	No Part Selected	☐	☐
P6	PicoBlade_5...	Molex_PicoBla...	No Part Selected	☐	☐
Q1,Q2	LTR	Phototransistor...	No Part Selected	☐	☐
Q3,Q4	LTR	Phototransistor...	No Part Selected	☐	☐
SW2	SS12D01G4	SS12D01G4	No Part Selected	☐	☐
U1,U9	MA302/MA730	MA302GQ-P	No Part Selected	☐	☐
U5,U6	ENCODER_...	ENC_CONN_F...	No Part Selected	☐	☐
U10,U2	ENCODER_...	ENC_CONN_V...	No Part Selected	☐	☐
U12	STLINK-V3M...	mod_stlink-v3...	No Part Selected	☐	☐

Please verify check the packages of selected parts before processing

For below not matched parts, try our Manual Parts Selection >					
D1,D3	OSI5FU3A11C	OSI5FU3A11C	No Part Selected	☐	Pre-order
D2,D4	OSI5FU3A11C	OSI5FU3A11C...	No Part Selected	☐	Pre-order
J2	Conn_01x04	PinSocket_1x0...	No Part Selected	☐	Pre-order
P1,P3	Red	motor_pin	No Part Selected	☐	Pre-order
P2,P4	Blk	motor_pin	No Part Selected	☐	Pre-order
P6	PicoBlade_5...	Molex_PicoBla...	No Part Selected	☐	Pre-order
Q1,Q2	LTR	Phototransistor...	No Part Selected	☐	Pre-order
Q3,Q4	LTR	Phototransistor...	No Part Selected	☐	Pre-order
SW2	SS12D01G4	SS12D01G4	No Part Selected	☐	Pre-order
U1,U9	MA302/MA730	MA302GQ-P	No Part Selected	☐	Pre-order
U5,U6	ENCODER_...	ENC_CONN_F...	No Part Selected	☐	Pre-order
U10,U2	ENCODER_...	ENC_CONN_V...	No Part Selected	☐	Pre-order
U12	STLINK-V3M...	mod_stlink-v3...	No Part Selected	☐	Pre-order

Please c

If JLCPCB is in stock, candidate parts will be automatically selected.

Parts data confirmation screen 2

Uploaded BOM Data			Review Matched Parts						
Top Designator	Comment	Footprint	Matched Part Detail	Qty	Source	Lib Type	Total Cost	Select ?	
C1,C11,C13,C16,...	0.1u	0402	CL05B104KO5NNNC C1525 16V 100nF X7R ±10% 0402 Multilayer Ceramic...	50	JLPCPB	Basic	¥9.5860 ?	☒	
C10,C12,C14,C1...	10u	0603	CL10A106KP8NNNC C19702 10V 10uF X5R ±10% 0603 Multilayer Ceramic ...	60	JLPCPB	Basic	¥49.8473 ?	☒	
C17	10n	0402	CL05B103KB5NNNC C15195 50V 10nF X7R ±10% 0402 Multilayer Ceramic ...	20	JLPCPB	Basic	¥3.8344 ?	☒	
C19	4.7u	0402	CL05A475MP5NRNC C23733 10V 4.7uF X5R ±20% 0402 Multilayer Ceramic ...	20	JLPCPB	Basic	¥16.6158 ?	☒	
D5	XL-1608UYC...	0603	XL-1608UYC-06 C965802 25mA 无色透明透镜 -30°C~+85°C 589nm~591nm...	20	JLPCPB	Extended	¥17.8939 ?	☒	
D7	XL-1608SYG...	0603	XL-1608SYG-06 C965805 25mA 2800K~3160K 无色透明透镜 -30°C~+85...	20	JLPCPB	Extended	¥17.5744 ?	☒	
D8,D9	XL-1608UBC...	0603	XL-1608UBC-04 C965807 25mA 无色透明透镜 -30°C~+85°C Positive post ...	20	JLPCPB	Extended	¥10.8642 ?	☒	
D10,D6	XL-1608SUR...	0603	XL-1608SURC-06 C965799 25mA 无色透明透镜 -30°C~+85°C Positive post ...	20	JLPCPB	Extended	¥9.5860 ?	☒	
J1,J4	BD125-10-A-...	PinHeader_2x0...	ZX-PZ1.27-2-5PZZ C7501289 Gold 500V 1A Straight Square Pins 1mm 10P 3...	0	JLPCPB	Extended	¥0.00 ?	☐	
L1,L2	47uH	Inductor	AMWPH4018S470MT C2846193 650mA 47uH ±20% 700mA SMD,4x4mm Powe...	10	JLPCPB	Extended	¥389.5119 ?	☒	
Q5,Q6,Q7,Q8	IRLML6344	SOT-23	IRLML6344 C5364313 30V 5A 1PCSNChannel SOT-23 MOSFETs RO...	24	JLPCPB	Extended	¥154.9101 ?	☒	
Q9	DMC2400UV-7	SOT-23	DMC2400UV-7 C177025	1	JLPCPB	Extended	¥22.8801 ?	☒	

Uncheck "2x5 pin header parts are different"

Parts data confirmation screen 3

Check for any incorrect parts.

Q1,Q2	LTR	Phototransistor...	No Part Selected	Search	Pre-order				
Q3,Q4	LTR	Phototransistor...	No Part Selected	Search	Pre-order				
SW2	SS12D01G4	SS12D01G4	No Part Selected	Search	Pre-order				
U1,U9	MA302/MA730	MA302GQ-P	No Part Selected	Search	Pre-order				
U5,U6	ENCODER_...	ENC_CONN_F...	No Part Selected	Search	Pre-order				
U10,U2	ENCODER_...	ENC_CONN_V...	No Part Selected	Search	Pre-order				
U12	STLINK-V3M...	mod_stlink-v3...	No Part Selected	Search	Pre-order				

Please carefully check the packages of selected parts before proceeding.

NEXT

Project has unselected parts

Some items have no parts selected yet, please confirm you have selected all the parts you want to assemble.

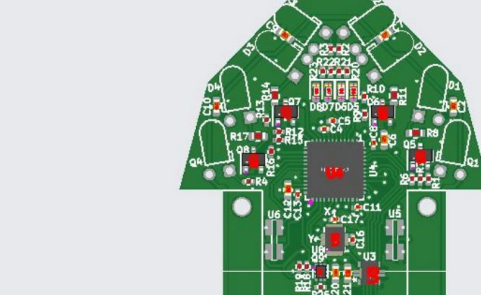
If you don't want to place those parts, just go ahead.

Do not place

Select parts

Selected part placement

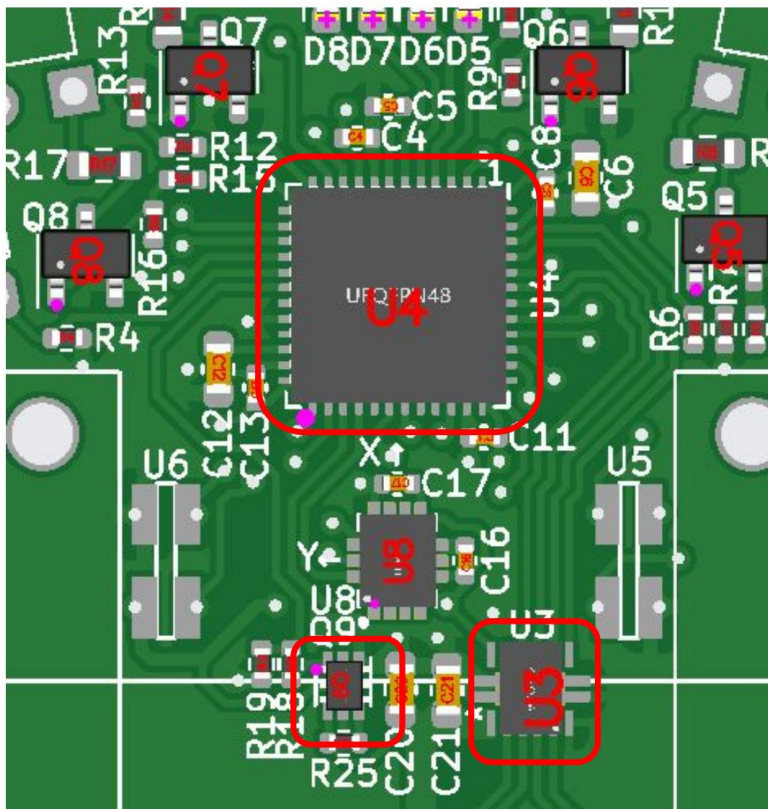
Parts placement confirmation and editing screen 1



Top Designator **JLPCB Part #** **Comment** **Footprint**

C1,C11,C13,C16...	C1525	0.1u	0402
C10,C12,C14,C1...	C19702	10u	0603
C17	C15195	10n	0402
C19	C23733	4.7u	0402
D5	C965802	XL-1608...	0603
D7	C965805	XL-1608S...	0603
D8,D9	C965807	XL-1608...	0603
D10,D6	C965799	XL-1608S...	0603
L1,L2	C2846193	47uH	Inductor
Q5,Q6,Q7,Q8	C5364313	IRLML63...	SOT-23
Q9	C177025	DMC240...	SOT-563
R1,R12,R15,R2...	C11702	1k	0402
R10,R13,R16,R7	C25741	100k	0402
R11,R14,R17,R8	C22859	10	0603
R18,R19,R25,R2...	C25744	10k	0402
R22	C25123	510	0402
SW1	C221676	KMR211...	KMR2xx...
U3	C191002	DRV8836...	DRV8836...

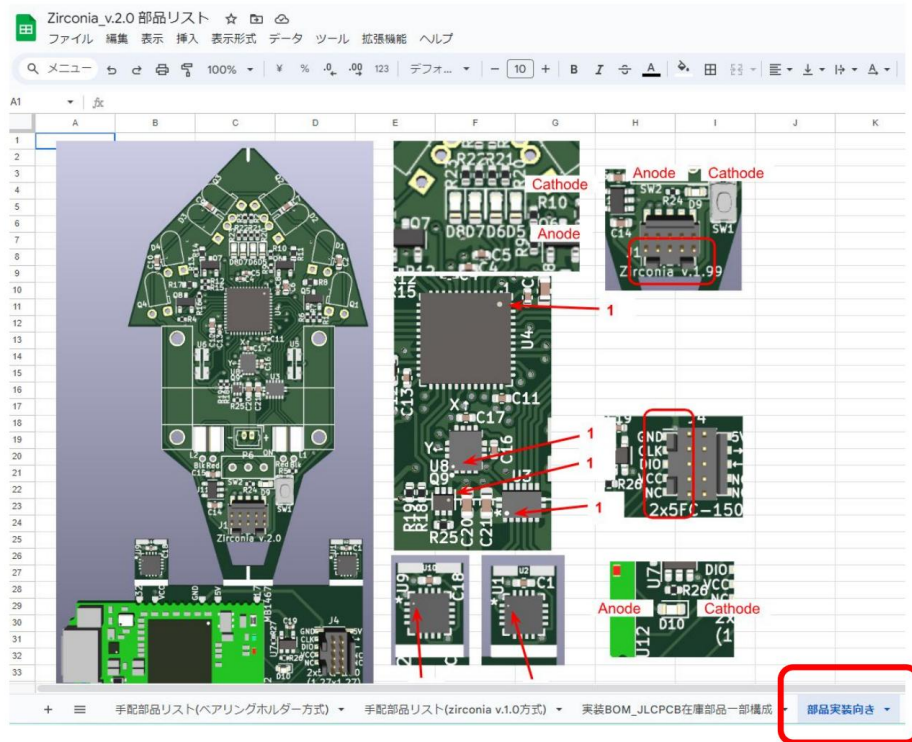
Parts placement confirmation & editing screen 2



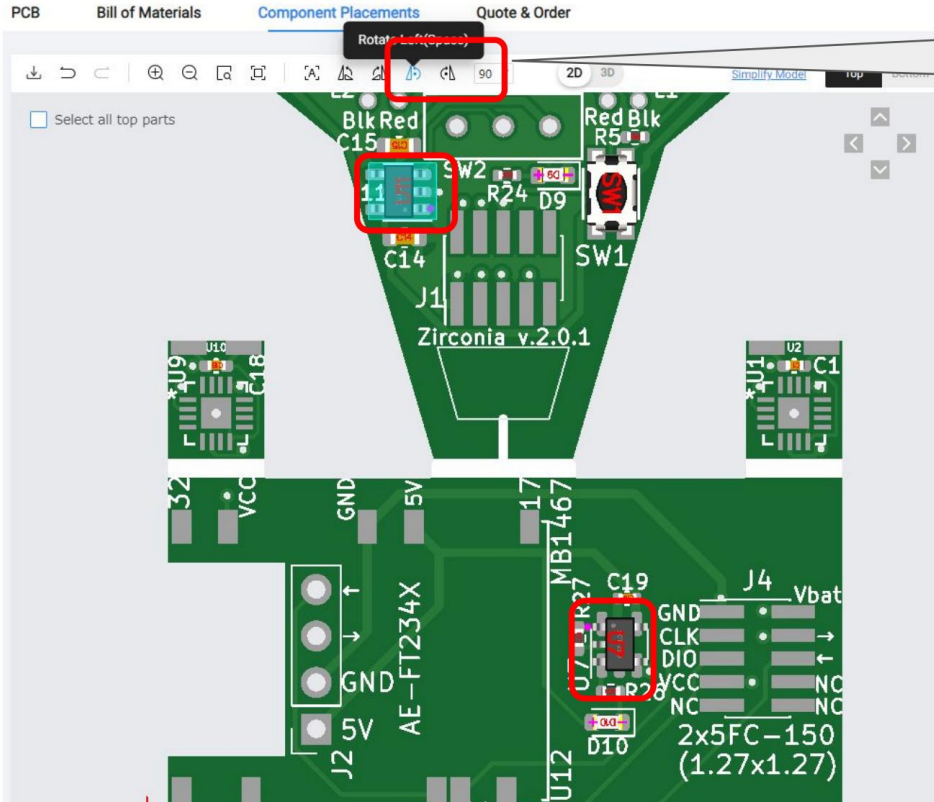
One pin is facing the wrong way.

Check the mounting direction of the parts list.

Correct the direction



Parts placement confirmation and editing screen 3



Orientation icon

The orientation of one pin is different, so correct the orientation of the component while checking the mounting orientation in the component list.

Parts placement confirmation and editing screen 4

Cost confirmation screen

Zirconia_v.2.0 [PCB](#) [Bill of Materials](#) [Component Placements](#) [Quote & Order](#)

Automatically saved, last updated on 22 June, 16:47

Charge Details

PCB Price	¥2462.01
Engineering fee:	¥639.07
Via Covering:	¥0.00
Panel:	¥1231.80
Surface Finish:	¥191.72
Board:	¥399.42

Standard PCBA Price	¥23289.24
Setup Fee:	¥3994.17
Stencil:	¥1255.77
Panel:	¥6847.61
Large Size:	¥0.00
Components(22 items):	¥6342.75
Feeders Loading fee:	¥4419.16
SMT Assembly	¥354.68
Packaging fee:	¥75.09

Build Time:

PCB: 3-4 days	¥0.00
Assembly: ☒ 2-3 days	¥0.00
☐ 1-2 days	¥7531.42

Total Price:

¥25751.25

Weight

402.82g

Product Description

Reserch\Education\DIY\Entertainment / R...

SAVE TO CART

Select the appropriate one
DIY "Robot"

I'm satisfied with the price and
put it in the basket

Basket contents

SHOPPING CART

All (3)

JLPCPB (3)

JLC3DP (0)

JLCMC (0)

☐ Item Qty Build Time Price ☐ JLPCPB (PCB/PCBA/Stencil)☒  Zirconia_v.2.0_Y22 Qty: 5 Build Time: 3-4 days Price: ¥2462.01 

PCB prototype:Y22-3893051A

Green, 0.6 Thickness, LeadFree HASL

[Product Details](#) | [Edit Order](#) Zirconia_v.2.0_Y22 Qty: 5 Build Time: 2-3 days Price: ¥23289.24

Standard PCBA: SMT02406221802700-38...

Assemble top side

[Product Details](#)

SUMMARY (2 items)

Subtotal **¥25751.25**Est. shipping date  2024-06-27

Order won't be shipped until all items are ready.

[Secure Checkout](#)[+ Add new item](#)

VISA

MasterCard

American Express

AMERICAN EXPRESS

DISCOVER

Diners Club

PayPal

Apple Pay

Google Pay

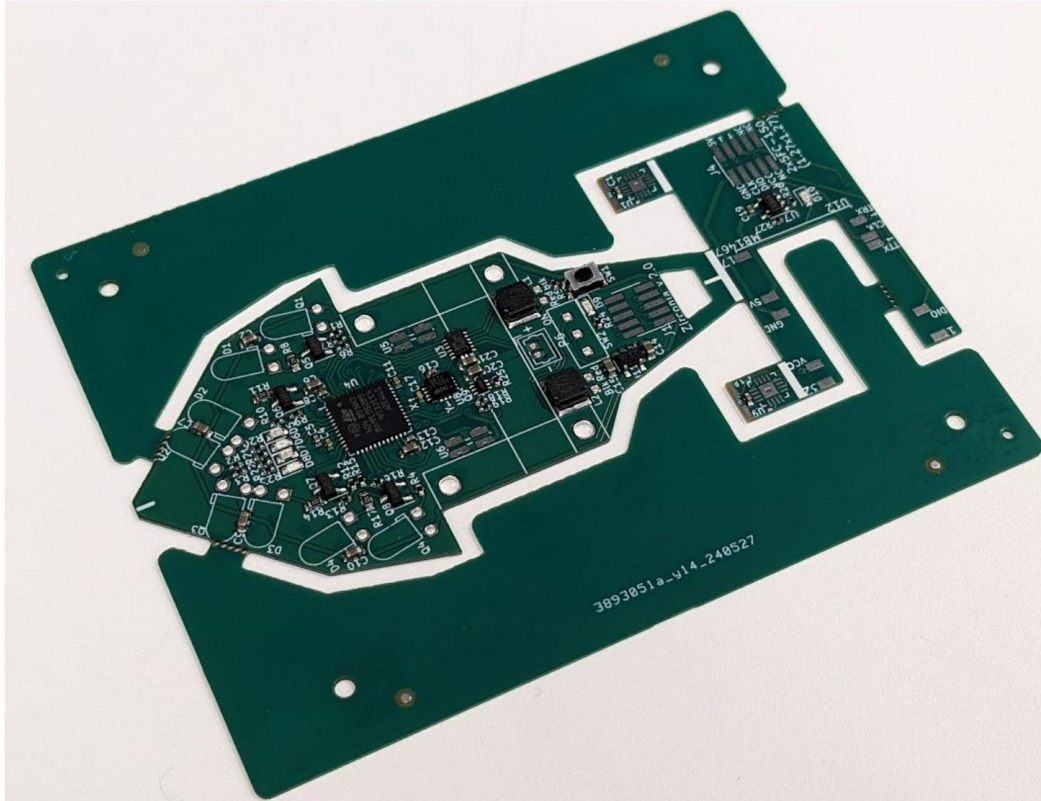
Bank Transfer

Payment method may vary by country.

 SSL ENCRYPTED PAYMENT

Check-out

Wait a week...



This is what the
circuit board looks like when it arrives.

I worked hard to install the remaining surface-mounted
components and lead components.